Tropical Meteorology

Chapter 3: Tropical Weather Systems and Variability

3.1 Tropical Weather Systems

Objective

- *To Know Thunderstorms*
- *To Identify The Life Cycle of a Thunderstorm*
- *To See the Easterly Waves*
- *To explain West African Disturbance Lines*

Key Characteristics of Tropical Weather Systems

- Weak horizontal temperature gradients
- Strong vertical motion
- High moisture content
- Dominance of latent heat energy

Thunderstorms

- Thunderstorms, also called cumulonimbus clouds, are tall, vertically developed clouds that produce lightning and thunder.
- Sometimes a thunderstorm produces gusty surface winds with heavy rain and hail.
- The majority of thunderstorms are not severe.
- The National Weather Service reserves the word “severe” for thunderstorms that have potential to threaten live and property from wind or hail.

Conti…..

- A thunderstorm is considered severe if:
  - Hail with diameter of three-quarter inch or larger, or
  - Wind damage or gusts of 50 knots (58mph) or greater, or A tornado

How Thunderstorms occurs?

How Thunderstorms occurs?

- The birth of a thunderstorm occurs when warm, humid air rises in a conditionally unstable environment.
- The initial process of formation of thunderstorm needs to start with air moving upward.
- This upward movement of air may be due to *unequal heating* of the surface, *effect of terrain*, or *lifting of warm air* along a frontal zone.
- Usually, several of these mechanisms may *work together* to generate severe thunderstorms.

Necessary Conditions for Formation

Thunderstorms require three key ingredients:

(a) Moisture

- High humidity in the lower troposphere
- Provides water vapor for cloud formation

(b) Instability

- Occurs when environmental lapse rate > moist adiabatic lapse rate
- Leads to buoyant rising air

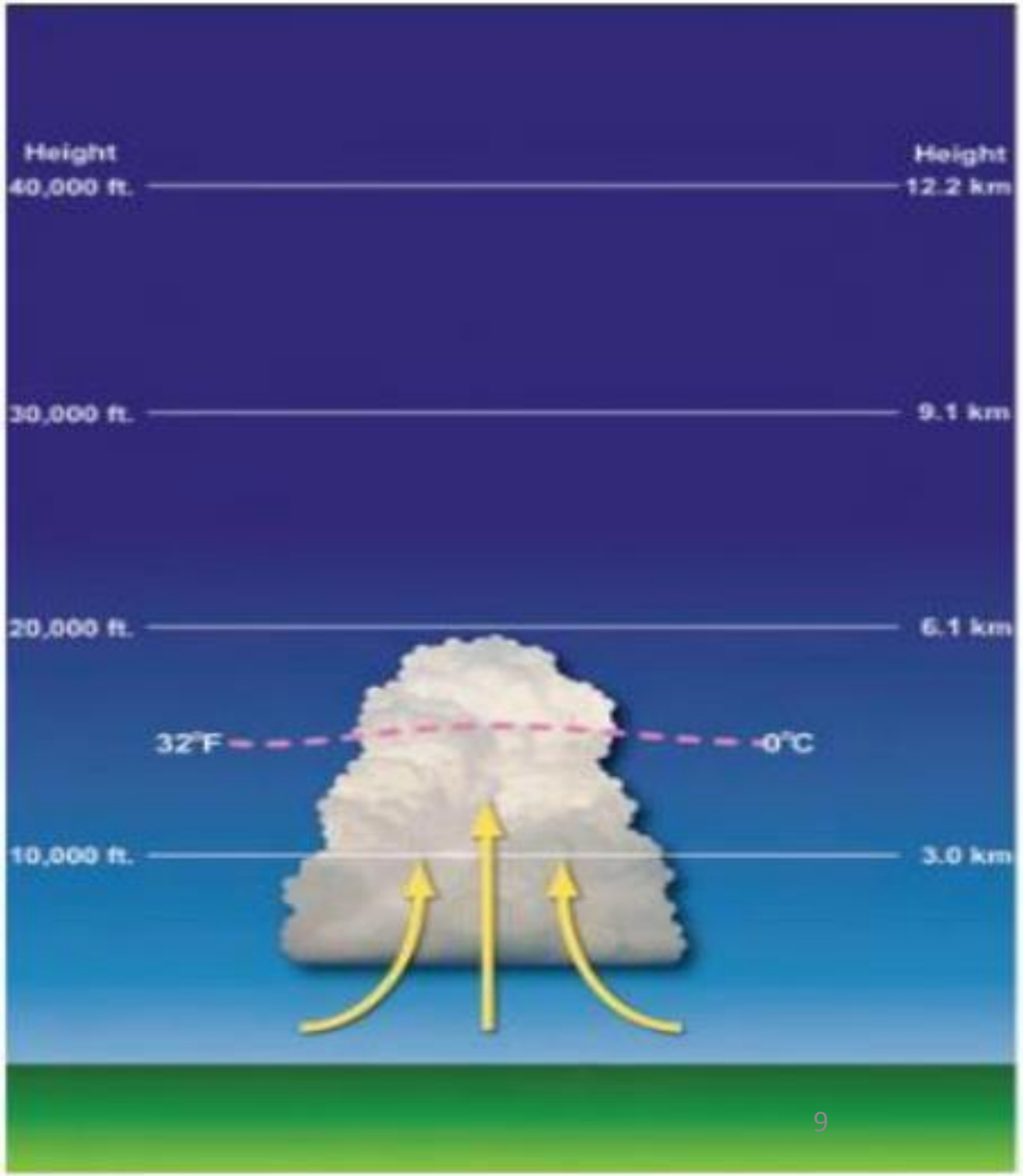
(c) Lifting Mechanism

- Surface heating (most common in tropics)
- Convergence (e.g., ITCZ)
- Orographic lifting

Life Cycle of a Thunderstorm

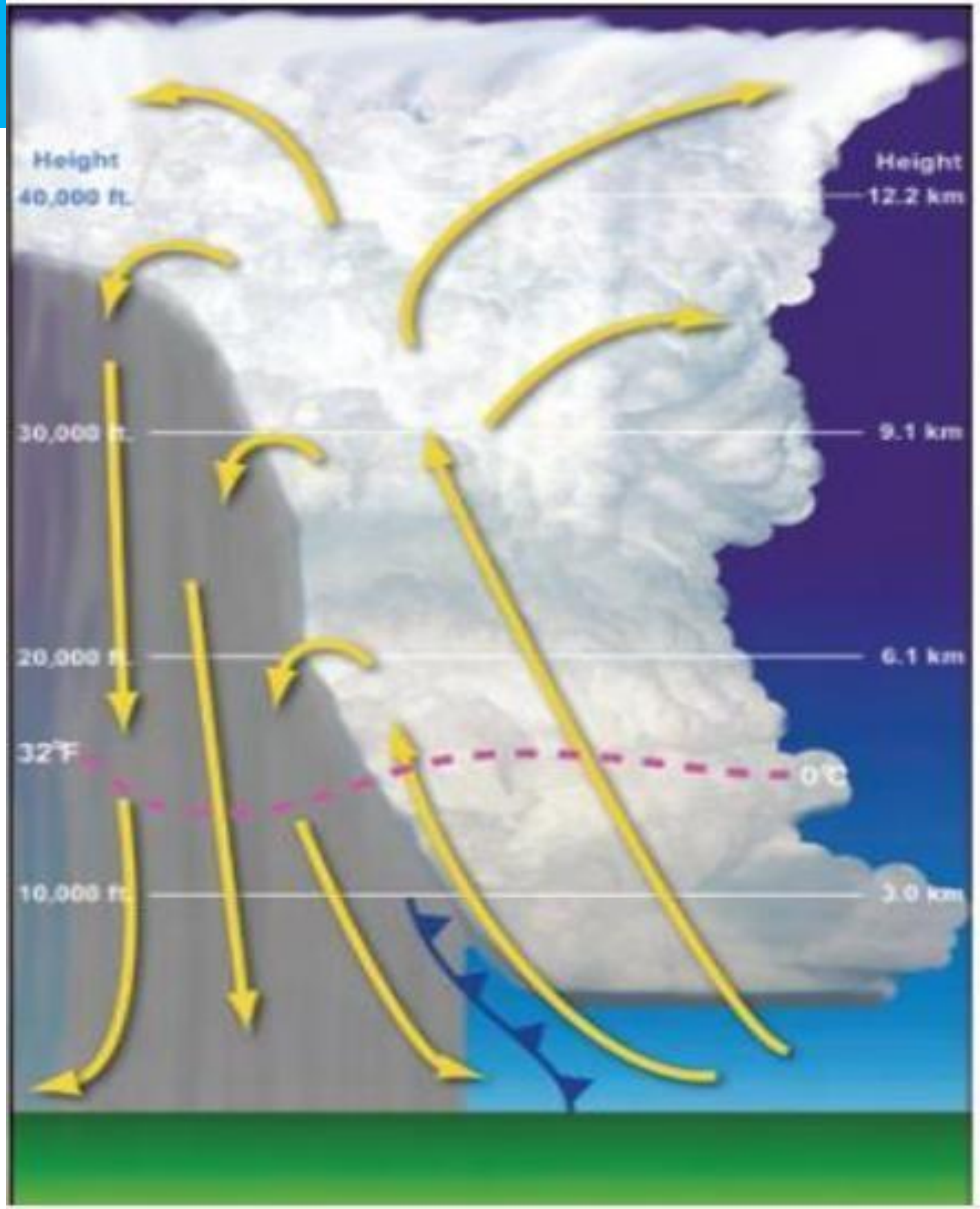
Life Cycle of a Thunderstorm

- ***Developing Stage (Cumulus Stage)***
  - Towering cumulus cloud indicates rising air.
  - Usually little if any rain during this stage.
  - Lasts about 10 minutes.
  - Occasional lightning.



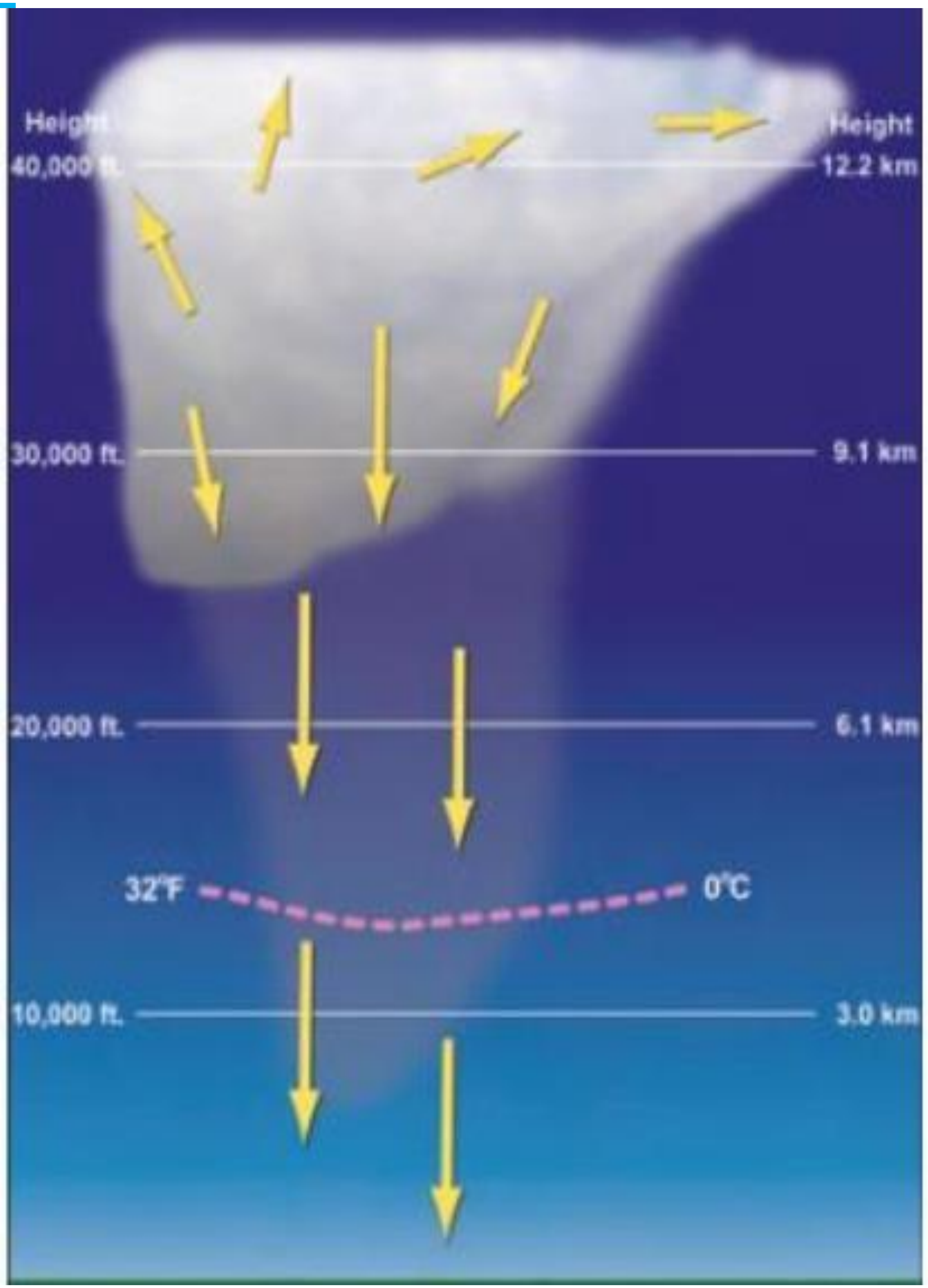
Mature Stage

- Most likely time for hail
- heavy rain
- frequent lightning
- strong winds
- tornadoes Storm occasionally has a black or dark green appearance
- Continuous Updrafts and Downdrafts
- Lasts an average of 10 to 20 minutes but some storms may last much longer



Dissipating Stage

- Downdrafts, downward flowing air, dominate th e storm
- Rainfall decreases in intensity
- Can still produce a burst of strong winds
- Lightning remains a danger



Types of Thunderstorms

Types of Thunderstorms

Thunderstorms are classified based on their structure, organization, and dynamics.

The four main types are:

- Single-Cell Thunderstorm
- Multi-Cell Cluster
- Squall Line
- Supercell Thunderstorm

Single-Cell (Pulse) Thunderstorms

- A single-cell thunderstorm is a simple, isolated storm consisting of one convective updraft.
- Characteristics
  - Short-lived: 30–60 minutes
  - Weak to moderate intensity
  - Forms in weak wind shear environments
  - Common in tropical and subtropical regions
  - Often occurs in the afternoon

Multi-Cell Cluster

- A multi-cell cluster is a group of thunderstorms at different stages of development.
- Characteristics
  - Several cells coexist
  - Each cell lasts ~30–60 min
  - Entire system lasts several hours
  - Moderate wind shear

Behavior

- New cells form as old ones decay
- Continuous regeneration

This makes the system longer lasting

than single-cell storms

Squall Line

- A squall line is a long, organized line of thunderstorms.
- Characteristics
  - Length: 100–1000 km
  - Moves as a single system
  - Strong wind shear required
  - Can last many hours to day

Tropical Example

- West African squall lines
- Common during monsoon seasons

Supercell Thunderstorm

- ❑ A supercell is a highly organized thunderstorm with a rotating updraft, called a Mesocyclone
- ❑ Characteristics
  - ✓ Long-lived (several hours)
  - ✓ Very strong wind shear required
  - ✓ Highly organized and powerful

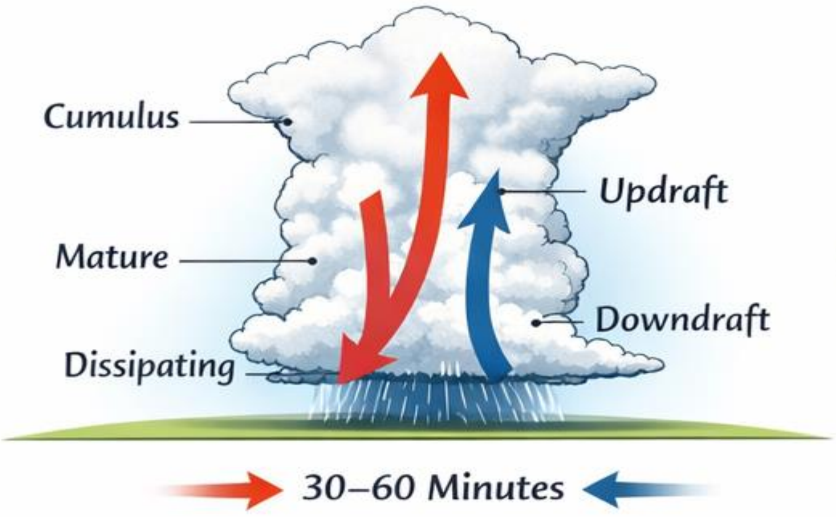
- ❑ Key Features
- Rotating updraft
- Can produce:
  - Tornadoes
  - Large hail
  - Severe winds

Comparison Summary Table

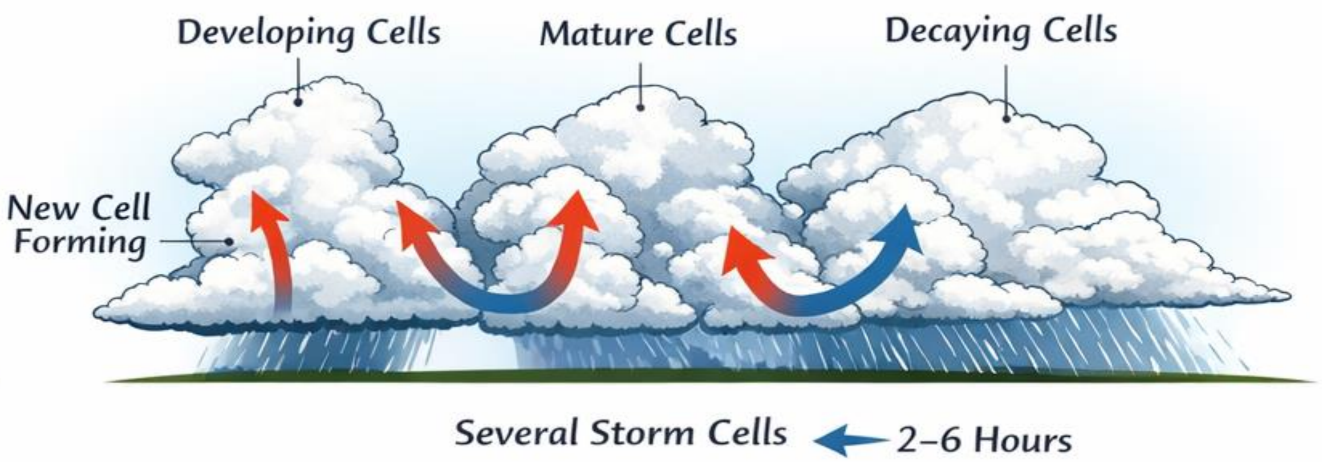
Feature	Single Cell	Multi-Cell	Squall Line	Supercell
Organization	Weak	Moderate	Strong (linear)	Very strong
Lifetime	<1 hour	Several hours	Many hours–days	Several hours
Wind Shear	Weak	Moderate	Strong	Very strong
Severity	Low	Moderate	High	Extreme
Special Feature	Simple life cycle	Regenerating cells	Line structure	Rotation (mesocyclone)

- In tropical meteorology, the most important thunderstorm types are single-cell, multi-cell clusters, and squall lines, while supercells are rare but possible under special conditions.

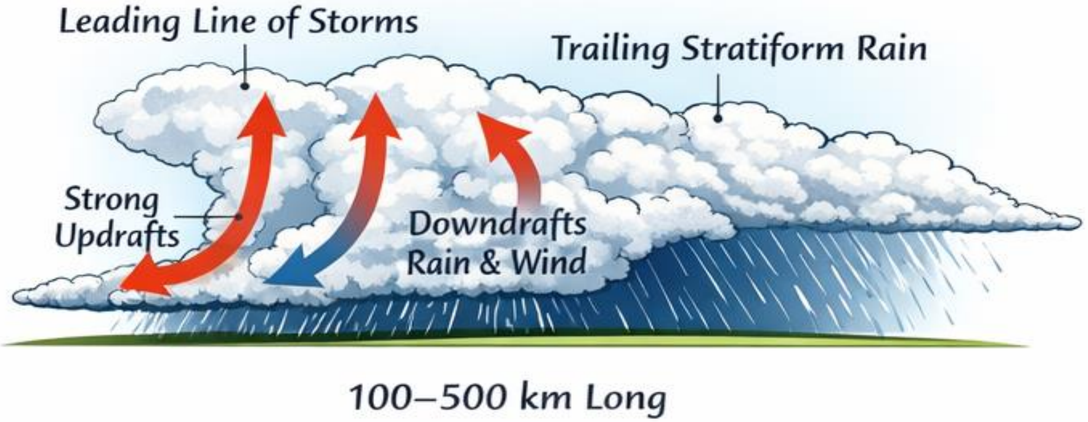
Single-Cell Thunderstorm



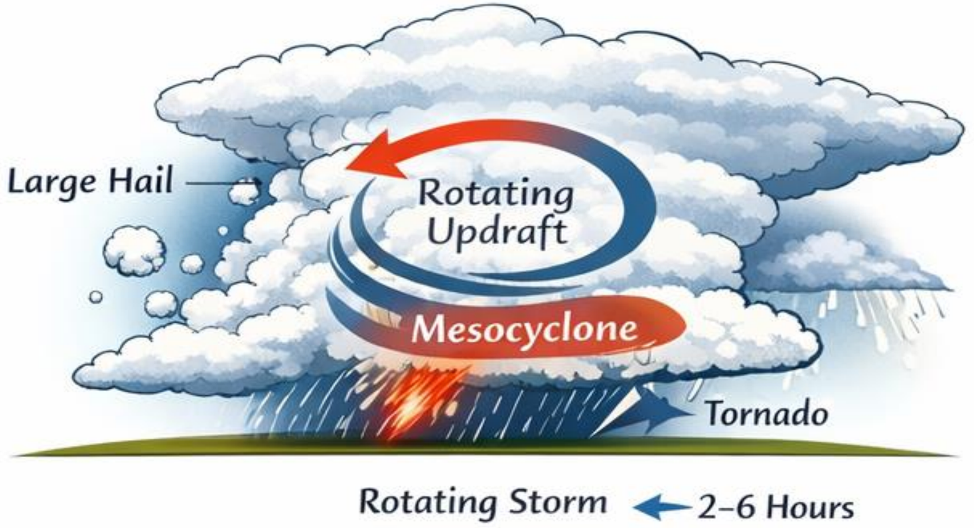
Multi-Cell Cluster



Squall Line



Supercell Thunderstorm



Tornadoes

Tornadoes

- A tornado is a violently rotatin column of air extending from a thunderstorm to the ground.
- The average tornado moves from southwest to northeast, but tornadoes have been known to move in any direction.
- The average forward speed is 30 mph but may vary from nearly stationary to 70 mph.
- Very small scale
- Extremely intense winds

conti.....

- The strongest tornadoes have rotating winds of more than 250 mph.
- Tornadoes can accompany tropical storms and hurricanes as they move onto land.
- Waterspouts are tornadoes which form over warm water. They can move onshore and cause damage to coastal areas.
- ***Enhanced Fujita Scale** is Used To measure the intensity of tornadoes*

When and Where Tornadoes Occur?

conti.....

- Tornadoes can occur at any time of the year.
- Tornadoes have occurred in every state, but they are most frequent east of the Rocky Mountains during the spring and summer months.
- In the southern states, peak tornado occurrence is March through May, while peak months in the northern states are during the late spring and summer.
- Tornadoes are most likely to occur between 3 and 9 p.m. but can happen at any time.

How Tornadoes Form

Before thunderstorms develop, winds change direction and increase in speed with altitude. This creates an invisible, horizontal spinning effect in the lower atmosphere.

Rising air within the thunderstorm updraft tilts the rotating air from horizontal to vertical.

An area of rotation, 2-6 miles wide, now extends through much of the storm. Most tornadoes form within this area of strong rotation.

Tornadoes Take Many Shapes and Sizes

Weak Tornadoes
- 88% of all tornadoes
- Less than 5% of tornado deaths
- Lifetime 1 – 10+ minutes
- Winds less than 110 mph

Strong Tornadoes
- 11% of all tornadoes
- Nearly 30% of all tornado deaths
- May last 20 minutes or longer
- Winds 110-205 mph

Powerful Tornadoes
- Less than 1% of all tornadoes
- 70% of all tornado deaths
- Lifetime can exceed 1 hour
- Winds greater than 205 mph

Tropical Cyclone

Tropical cyclone

- A *tropical cyclone* is an especially intense low-pressure system of counter clockwise circulation in tropical latitudes, away from the equator in Northern Hemisphere.
- The circulation of tropical cyclone in case of Southern Hemisphere is clockwise.
- A tropical cyclone is a rapidly rotating storm system, and that produce heavy rain.
- A tropical cyclone is a large, rotating, warm-core low-pressure system that forms over warm tropical oceans.

Tropical cyclones (TCs) are called

- Hurricanes in North Atlantic
- Typhoons in North West Pacific
- Williwillies in South Pacific
- Cyclones in Indian Ocean
- Bagyo in Philippine seas
- Asifat in Gulf of Arabia.

Formation and Intensification TC

Essential *Conditions for Formation of TC*

- Sea surface temperature ≥ 26–27°C
- Latitude > 5° (Coriolis force needed)
- High humidity

Structure of a Tropical Cyclone

(a) Eye

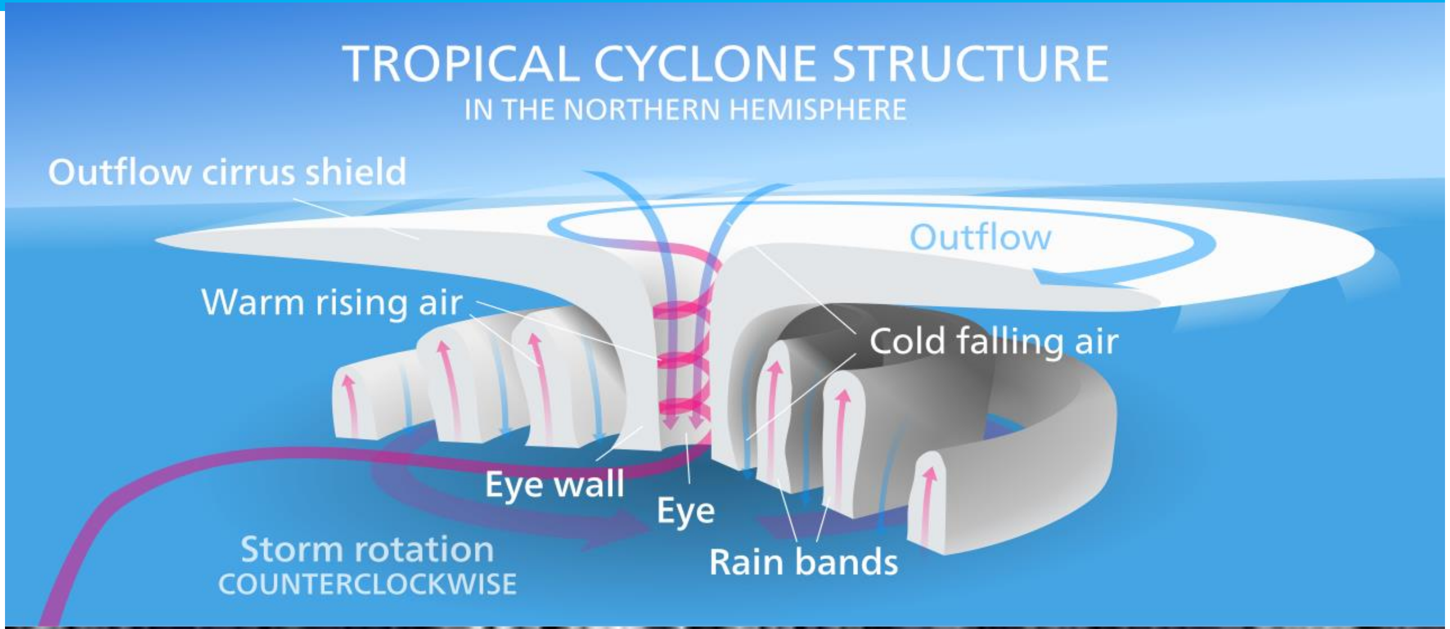
- Calm, clear center
- Lowest pressure

(b) Eyewall

- Strongest winds
- Heaviest rainfall

(c) Spiral Rainbands

- Bands of clouds and rain



Decaying Stage of TC

- A tropical cyclone decays for several reasons:
- *It moves inland*, where the cyclone's inflow is no longer made humid by the surface, and surface friction is greater than over the sea, It moves to an area of the sea which is too cool,
- The TC remains stationary, steadily cooling the sea beneath, until it is too cold to supply the required heat,
- It shifts to higher latitudes, where the ground's vorticity is greater, reducing the cyclone's spin relative to the ground,
- The upper divergence (which sucks the central core upwards) becomes detached by the tug of upper winds, allowing upper westerlies to encroach into the system.

Easterly Waves (A tropical wave, tropical easterly wave, and African easterly wave)

- Easterly waves are westward-moving disturbances in the tropical atmosphere, embedded in the trade winds, especially over Africa and the Atlantic.
- Basic Characteristics
  - ✓ Move from east → west
  - ✓ Typical speed: 5–10 m/s
  - ✓ Causing areas of cloudiness and thunderstorms
  - ✓ Period: 3–5 days
  - ✓ Most active between 5°N–20°N

Structure of Easterly Waves

(a) Wave Trough

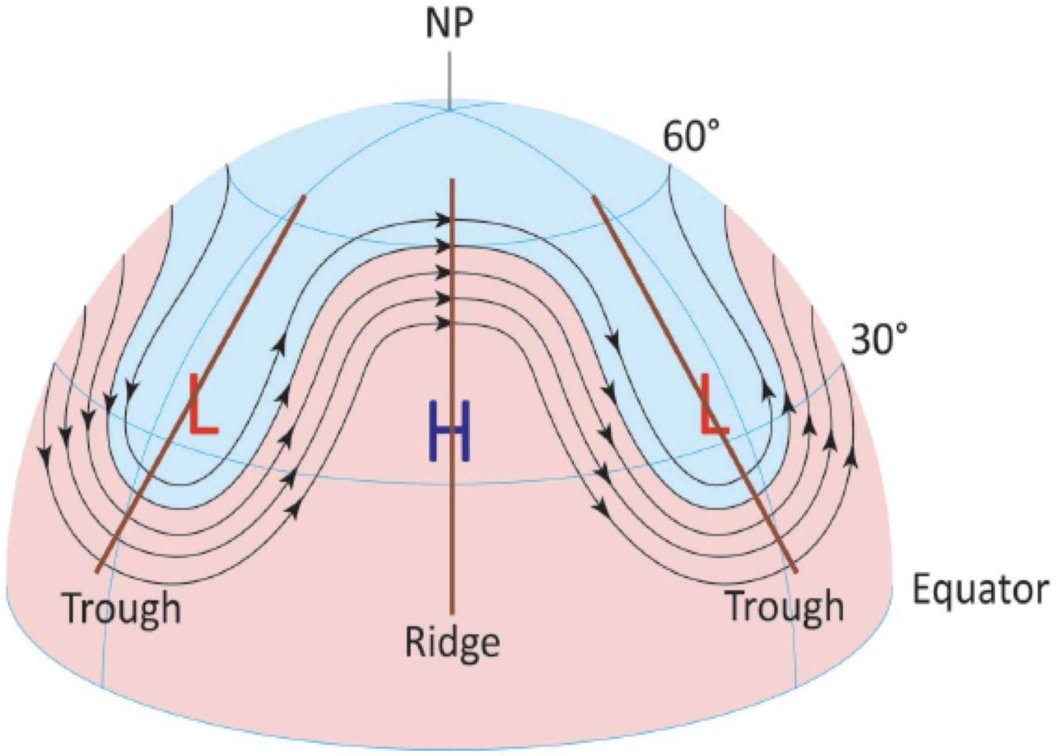
- Region of low pressure
- Convergence and rising motion
- Cloud formation and rainfall

Vertical Structure

- Strongest at mid-levels (~600–700 mb)
- Associated with the African Easterly Jet

(b) Wave Ridge

- Region of high pressure
- Divergence and sinking motion
- Clear weather



Tropi gure 9A-1. Wave pattern in upper-air westerlies.

Formation Mechanism

- Caused by instability of the African Easterly Jet
- Temperature contrast between:
  - Hot Sahara Desert
  - Cooler Gulf of Guinea

This creates barotropic and baroclinic instability

Weather Associated

- Cloud clusters
- Thunderstorms
- Heavy rainfall

Conti……

- Easterly waves originate from instabilities in the African Easterly Jet caused by temperature contrasts between the Sahara and the Gulf of Guinea.
- These waves move westward and can develop into hurricanes over the Atlantic if environmental conditions are favorable.

Conti……

- The African Easterly Jet is a mid-level wind current formed by temperature contrasts over Africa, while easterly waves are disturbances that develop within this jet due to instability and are responsible for much of the tropical weather activity.

- Easterly waves form because the AEJ becomes unstable

West African Disturbance Lines (Squall Lines)

- These are long, organized lines of thunderstorms that move westward across West Africa, often producing intense rainfall, strong winds, and lightning
- Location
  - Form over:
    - Sudan region
    - Sahel
  - Move toward:
    - West African coast
    - Atlantic Ocean

Where and When Do They Occur?

- ❑ Region: Mainly Sahel and Sudanian zone of West Africa
- ❑ Countries: Senegal → Mali → Niger → Nigeria
- ❑ Season: Rainy season (May–September)
- ❑ Movement: East → West

Conti.

- West African disturbance lines (squall lines) are organized, fast-moving lines of thunderstorms formed by the interaction of easterly waves, temperature contrasts, and wind shear, and they are a major source of rainfall and severe weather in the Sahel region.

Characteristics

- Move westward
- Speed: up to 10° longitude/day
- Length: 100–1000 km
- Lifetime: several hours to days

Importance

- Major source of rainfall in West Africa
- Can cause:
  - Flooding
  - Strong winds
- Key part of Mesoscale Convective Systems (MCS)

Step-by-step process

AEJ forms

↓

Instability develops

↓

Easterly waves form

↓

Waves organize convection

↓

Squall lines develop

Feature	AEJ	Easterly Waves	Squall Lines
Type	Wind system	Wave/disturbance	Storm system
Scale	Largest	Medium	Smallest
Height	Mid-level	Mid-level	Surface → upper
Structure	Continuous	Oscillating	Line of storms
Weather	Indirect	Some rain	Heavy rain, wind

….END

Thanks !

Objective

3.1 Tropical Weather Systems

- *To define tropical upper tropospheric troughs*
- *To Know Tropical Easterly jet stream*
- *To be aware of East African low level jet*
- *To describe the Somali jet*
- *To Define heat lows*

Tropical Upper Tropospheric Troughs (TUTTs)

TUTTs

- A Tropical Upper Tropospheric Trough (TUTT) is a cold-core trough located in the upper troposphere (200–100 mb) in tropical and subtropical regions.
- Location

- Tropical oceans (Atlantic, Pacific)
- Subtropical regions

Characteristics

- Cold-core system
- Located at high altitude
- Associated with low pressure aloft

Weather Effects

Can Enhance Convection

- If aligned with:
  - Moist air
  - Surface disturbances

Leads to:

- Thunderstorms
- Heavy rainfall

Can Suppress Convection:

- If:
- Strong subsidence dominates
- Increases wind shear

TUTTs

- Upper-level low-pressure systems in the tropical troposphere.
- Can enhance or inhibit tropical cyclone development depending on their interaction.
- Characterized by strong winds aloft and divergence that aids convection.
- Commonly found in the Atlantic and Pacific hurricane regions.

Tropical Easterly Jet Stream (TEJ)

Tropical Easterly Jet Stream (TEJ)

- A high-altitude wind current flowing east to west in the tropics.
- Peaks during the monsoon season, contributing to the transport of moisture.
- Influences the Indian monsoon and tropical cyclone formation.
- Originates over the Tibetan Plateau and flows toward the African region.

Conti.

Enhances convection

- Provides upper-level outflow
- Supports:
  - Deep clouds
  - Heavy rainfall

East African Low-Level Jet (EALLJ)

East African Low-Level Jet (EALLJ)

- A near-surface wind current flowing across East Africa.
- Plays a key role in transporting moisture from the Indian Ocean inland(East Africa).
- Contributes to the seasonal rainfall patterns in East Africa.
- regional moisture transport & convergence.

Conti…

Characteristics

- Direction: mostly easterly or southeasterly
- Transports:
  - Moisture
  - Momentum

Role

- Influences:
  - Ethiopian rainfall
  - Seasonal variability
- Interacts with:
  - Topography (highlands)

The Somali Jet

The Somali Jet

- A strong, low-level jet stream over the western Indian Ocean.
- Key contributor to the Indian monsoon by transporting moisture inland.
- Develops due to the temperature gradient between the land and ocean.
- Strengthens during the summer monsoon season, impacting East Africa and South Asia.

The Somali Jet Vs. *East African Low-Level Jet*

The Somali Jet

- ❑ large-scale, strong, cross-equatorial flow
- ❑ feeds the entire East African monsoon system

East African Low-Level Jet

- ❑ more regional and localized
- ❑ channels moisture into Ethiopia, Kenya, South Sudan
- ❑ helps trigger convection and rainfall

Heat Lows

- Intense low-pressure systems that form over hot land surfaces.
- Result from extreme surface heating and associated rising air.
- Drive regional wind circulations, influencing monsoon systems.
- Common in arid and semi-arid regions like the Sahara Desert.

Conti.

Role

✓ Drives circulation

- Creates pressure gradients
- Initiates:
    - Monsoon flow
    - Jet formation

Connection

- Helps form:
    - African Easterly Jet
- Influences:
    - Easterly waves
    - Squall lines

Level	System	Role
Upper	TUTT	Enhances/suppresses storms
Upper	TEJ	Supports convection
Mid	AEJ	Generates waves
Low	Somali Jet	Moisture supply
Surface	Heat Low	Drives circulation
Regional	East African LLJ	Local rainfall control

Upper Level

TUTT / TEJ → enhance or suppress storms

Storm Level

Squall Lines → produce rainfall

Disturbance

Easterly Waves → organize convection

Mid Level

AEJ → creates instability

Low Level

Somali Jet/East African LLJ → brings moisture

Surface

Heat Low → drives circulation

….END

Thanks !

Objective

3.1 Tropical Weather Systems

- *To know Inter-tropical Convergence Zone (ITCZ)*
- *To Know observational aspects of the ITCZ*
- *To see ITCZ theory*
- *To Know trade wind regimes*
- *To explain trade wind inversion*
- *To describe lighting*

Inter-Tropical Convergence Zone (ITCZ)

What is the ITCZ?

- Zone of low pressure near the equator
- Convergence of trade winds
- Strong convection and rainfall
- Key component of Hadley circulation
- The ITCZ exists over both oceans and continents

Observational Aspects of the ITCZ

Location

• Generally found between ~20°N and 20°S (tropical belt), but shifts seasonally.

• Follows the thermal equator (zone of maximum heating), not the geographic equator.

Seasonal movement

• Boreal summer (June–Sept): shifts northward (up to 15–20°N over continents like Africa and India)

• Boreal winter (Dec–Feb): shifts southward

Conti……………

Cloud and Precipitation Features

- Deep cumulonimbus clouds (Cb)
- Persistent thunderstorms
- Heavy rainfall zones (rainforests, monsoon regions)

Satellite signature:

- Bright white cloud band in infrared imagery
- Organized cloud clusters and mesoscale convective systems

Wind Structure

- Weak surface winds near center ("doldrums")
- Convergence of:
  - NE trade winds (Northern Hemisphere)
  - SE trade winds (Southern Hemisphere)

Intertropical Convergence Zone (ITCZ) Theory

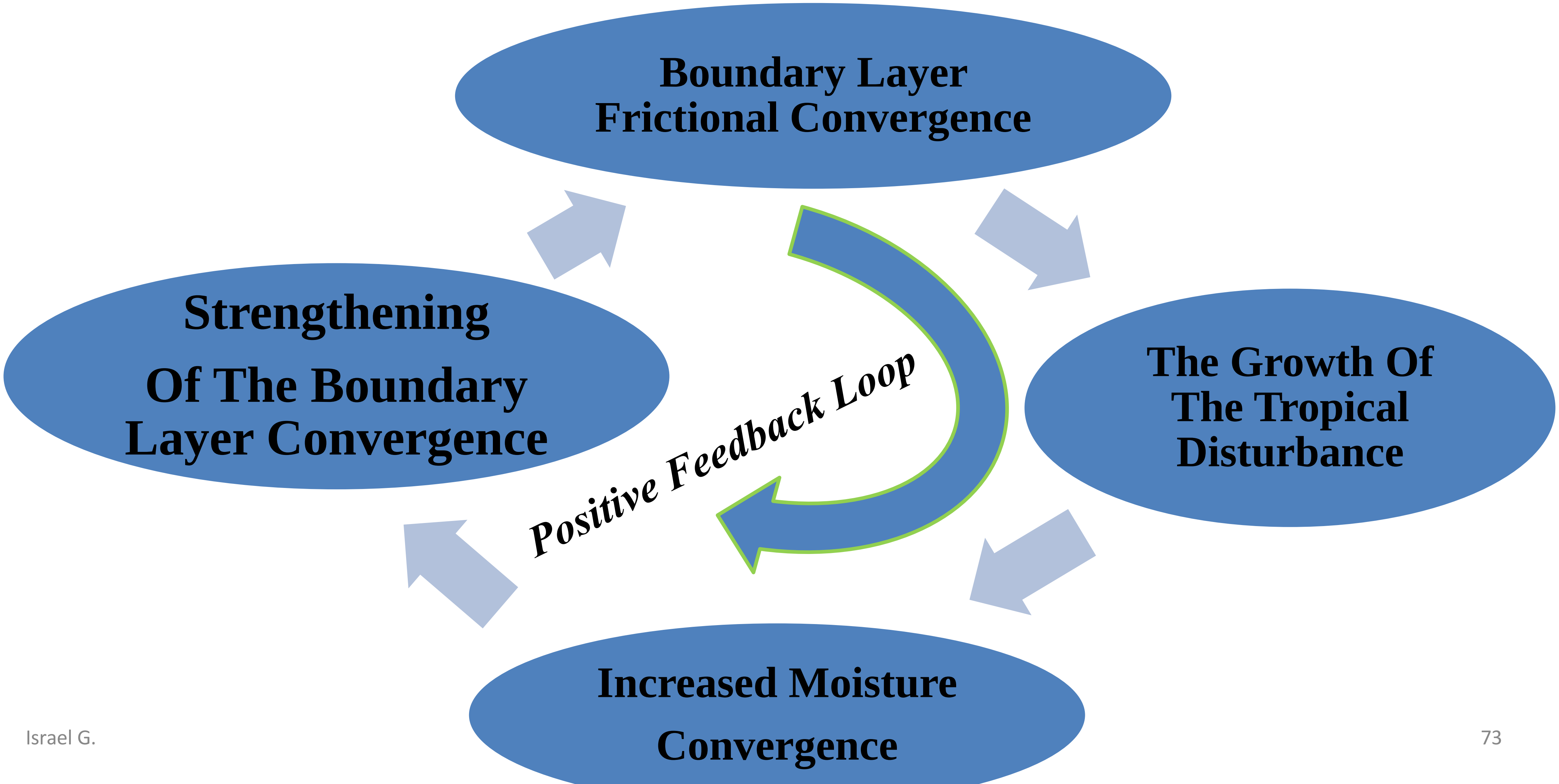
Early Theory – CISK

Conditional Instability of the Second Kind (CISK)

- Proposed by Charney et al. (1971)

✓ From this theory, the latitudinal location of the ITCZ is determined by the balance of two forces: the *rotation of the Earth* and the *abundance of moisture in the boundary layer.*

CISK Mechanism



Early Theory of ITCZ: CISK (Conditional Instability of the Second Kind)

- The CISK theory (Charney et al., 1971) explains the ITCZ as a result of positive feedback between boundary layer convergence and deep convection.
- It was the earliest theory of ITCZ

CISK Interpretation of ITCZ Location

CISK suggests ITCZ position is controlled by:

❖ **Earth's rotation**

• Coriolis force favors organized convergence

• Can shift ITCZ away from equator

❖ **Moisture availability**

•More moisture → stronger convection → ITCZ forms

ITCZ position = balance between

- Coriolis force → favors off-equator
- Moisture → favors equator

Problem with CISK Theory (Aqua-Planet Paradox)

Aqua-planet experiments

- Entire Earth covered by ocean
- No continents
- Uniform or near-uniform SST and radiation

But

ITCZ still forms near equator

Therefore, CISK is incomplete

Modern Theory: ITCZ from Rotational Balance (Chao & Chen, 2004)

- The Core idea is ITCZ location is controlled by two competing rotational effects
- ***(A) Coriolis "Spring Effect" (Inertial Stability)***

  At equator ($f = 0$):
  •No resistance → easy convergence
  ✓ ITCZ prefers equator

- ***(B) Wind–Evaporation Feedback***
  - Rotation → stronger winds
  - Strong winds → more evaporation
  - More evaporation → stronger convection
  - Pushes ITCZ **away from equator**

The ITCZ is the band of bright white clouds that cuts across the center of the image above.

Source: Wikicommons/NOAA

Conti……………

- Since water has a higher heat capacity than land (the ocean heats up more slowly than the land), the ITCZ propagates poleward more prominently over land than over water, and over the Northern Hemisphere than over the Southern Hemisphere.
- The position of the ITCZ varies with the seasons, and lags behind the sun's relative position above the Earth's surface by about 1 to 2 months, and correlates generally to the thermal equator.

Conti…

- In July and August, over the Atlantic and Pacific, the ITCZ is between 5 and 15 degrees north of the Equator, but further north over the land masses of Africa and Asia.
- In eastern Asia, the ITCZ may propagate up to 30 degrees north of the Equator.

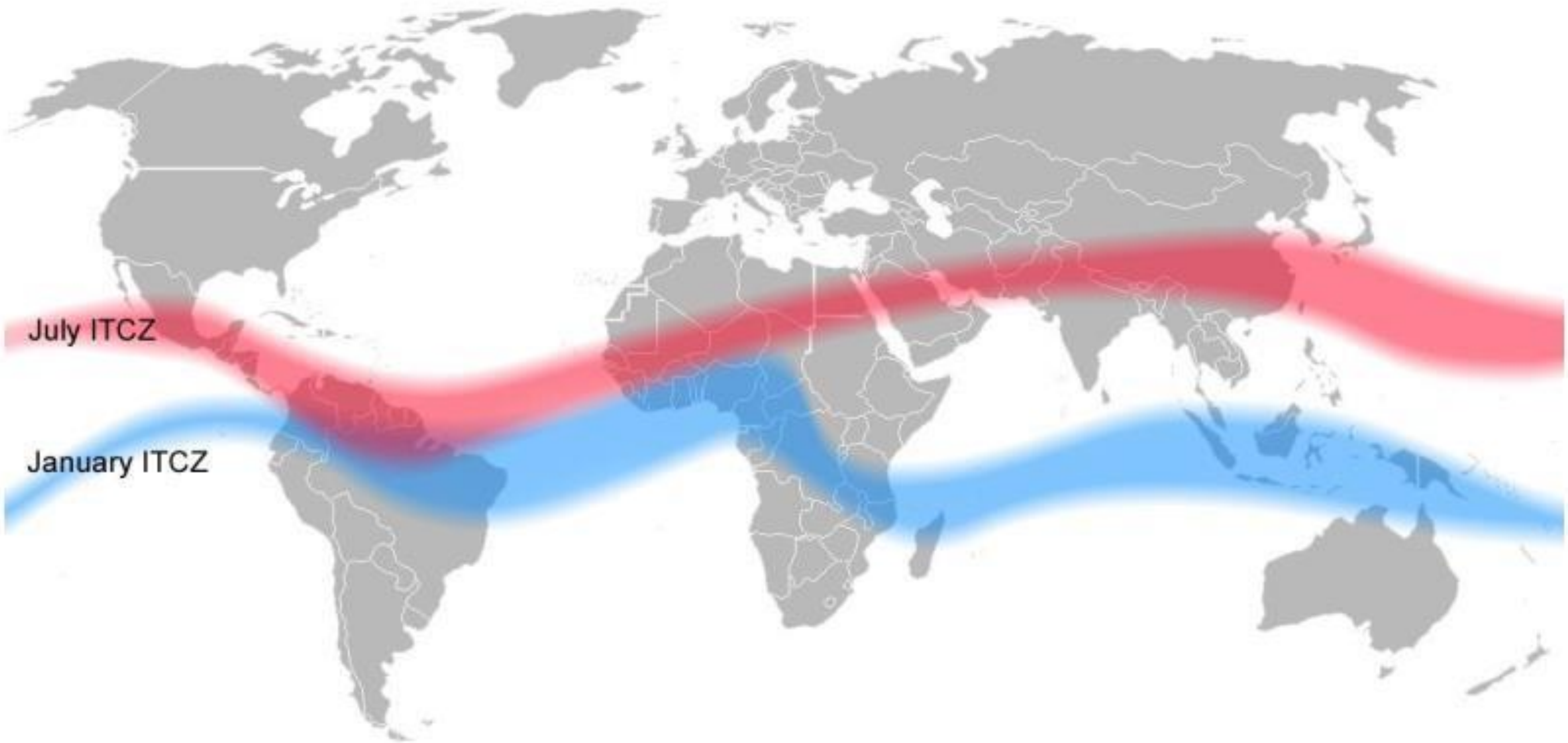
Conti…………….

- In January, over the Atlantic, the ITCZ generally extends much further south over South America, Southern Africa, and Australia.
- During this time period, the Southern Hemisphere is tilted towards the Sun and receives higher inputs of shortwave radiation.

Conti……………

- Note that the line representing the ITCZ is not straight and parallel to the lines of latitude. Bends in the line occur because of the differential heating characteristics of land and water.
- Over the continents of Africa, South America, and Australia, these bends are toward the South Pole. This phenomenon occurs because land heats up faster than ocean.

Seasonal location of ITCZ



Trade Wind Regimes

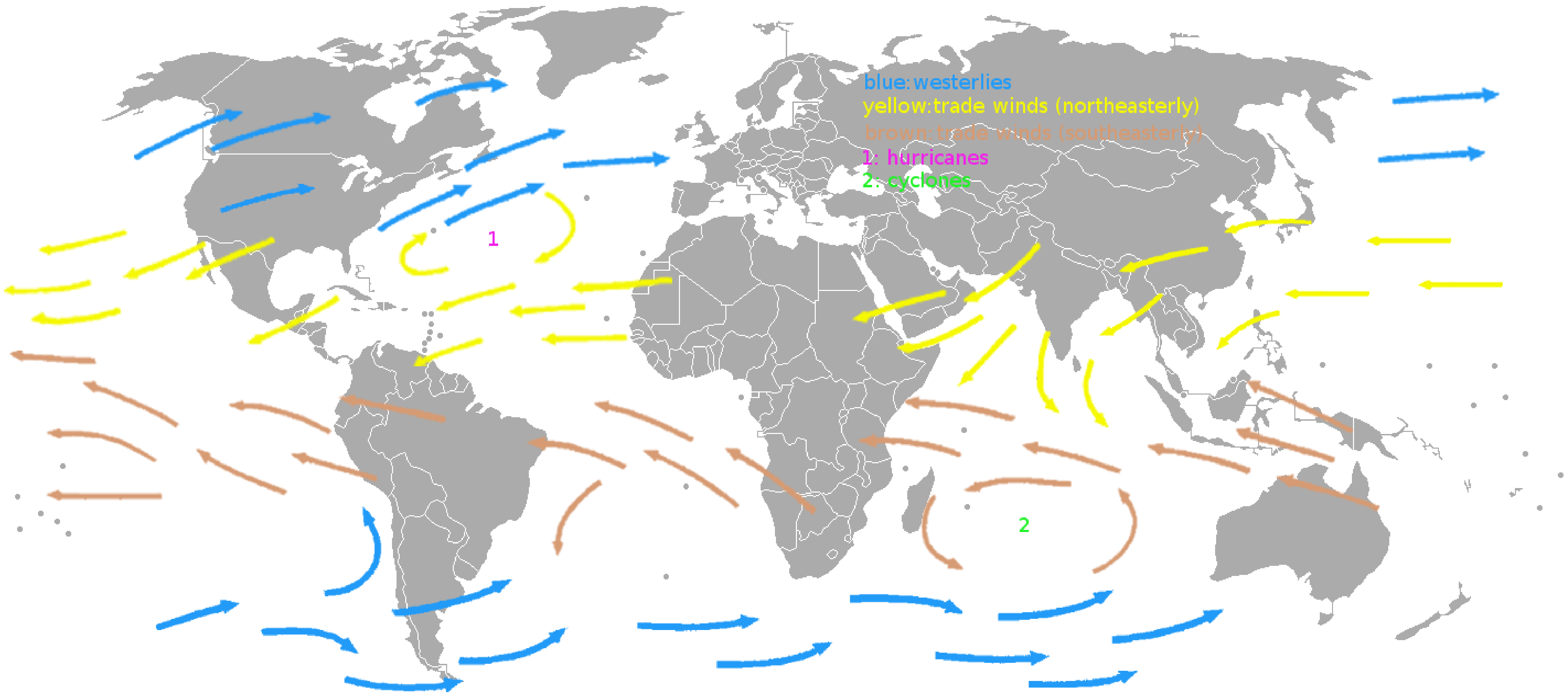
Trade Winds

- Trade winds are persistent easterly surface winds that blow:
  - From subtropical high-pressure regions (~30°)
  - Toward the equatorial low-pressure region (ITCZ)
- They blow from east to west → that's why they are called *easterlies*

Why they form

- Air rises at ITCZ
- Moves poleward aloft
- Sinks at ~30° (subtropical highs)
- Returns to equator at surface
- Coriolis deflects the flow

Trade Wind Regimes



Trade Wind Regimes

- A trade wind regime refers to the typical atmospheric structure and weather conditions in regions dominated by trade winds.
- Main Features

Strong, consistent easterlies

Shallow cumulus clouds, Sometimes stratocumulus

Tropical oceans (e.g., Atlantic, Pacific trade wind belts)

Trade Wind Inversion

Trade Wind Inversion

- A trade wind inversion is a layer in the lower atmosphere where temperature increases with height, which caps vertical motion.

How it forms

1. Air descends from subtropical highs
2. Descending air warms adiabatically
3. Warm, dry air sits above cooler moist air
4. Creates an inversion layer

- Prevents clouds from growing vertically

Conti……………

Feature	Trade Winds	Trade Wind Regime	Trade Wind Inversion
Type	Wind system	Climate pattern	Vertical structure
Role	Transport air	Define weather	Control convection
Convection	Weak	Shallow	Suppressed
Location	Tropics	Tropical oceans	~800–900 hPa

Lightning

Lightning

- Lightning is a sudden, powerful electrical discharge in the atmosphere, usually occurring within or between thunderstorm clouds, or between a cloud and the ground.
- It produces:
    - ✓ A bright flash (light)
    - ✓ Very high temperature (~30,000°C)
    - ✓ Thunder (sound from rapid air expansion)

Where does it occur?

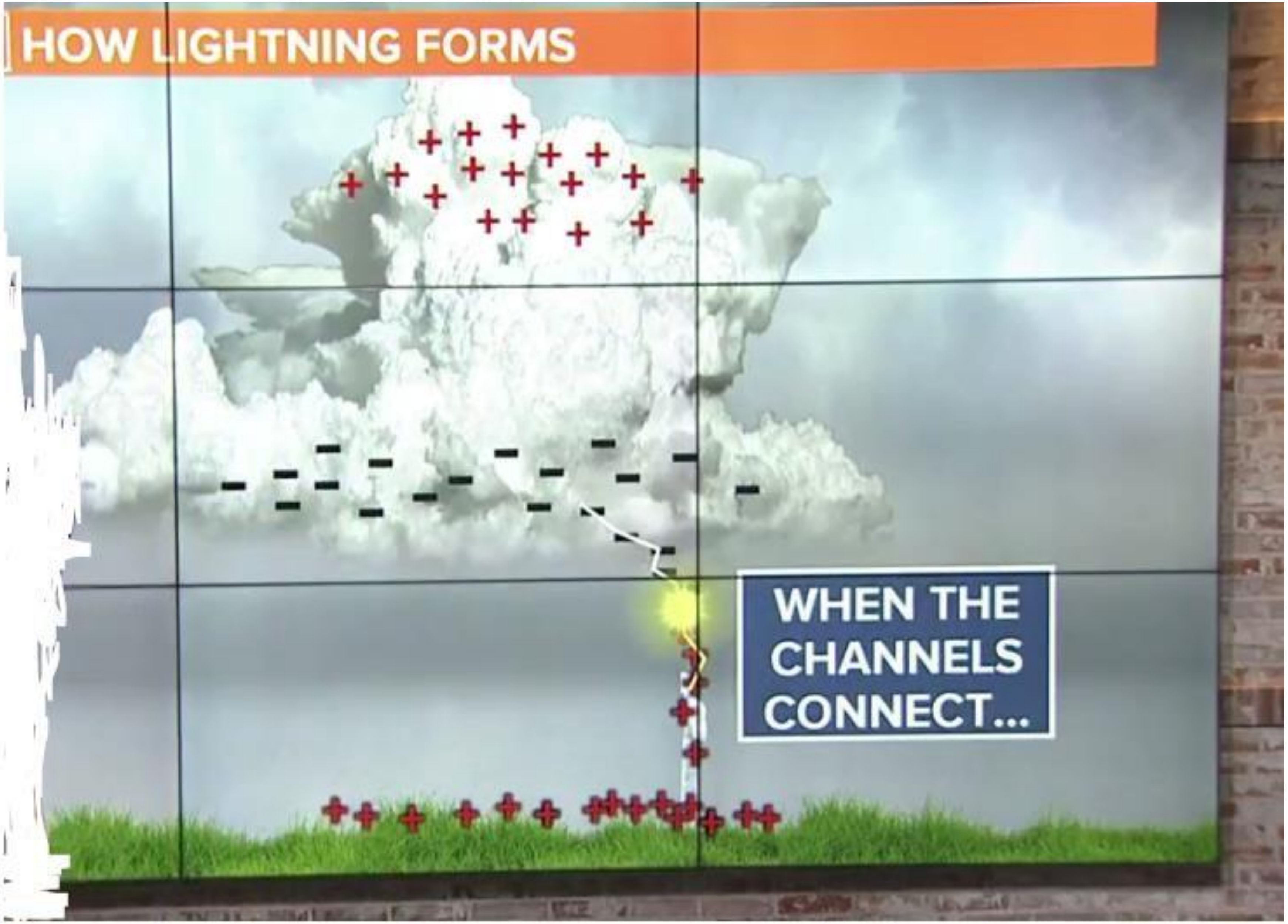
- Inside cumulonimbus (thunderstorm) clouds Between:

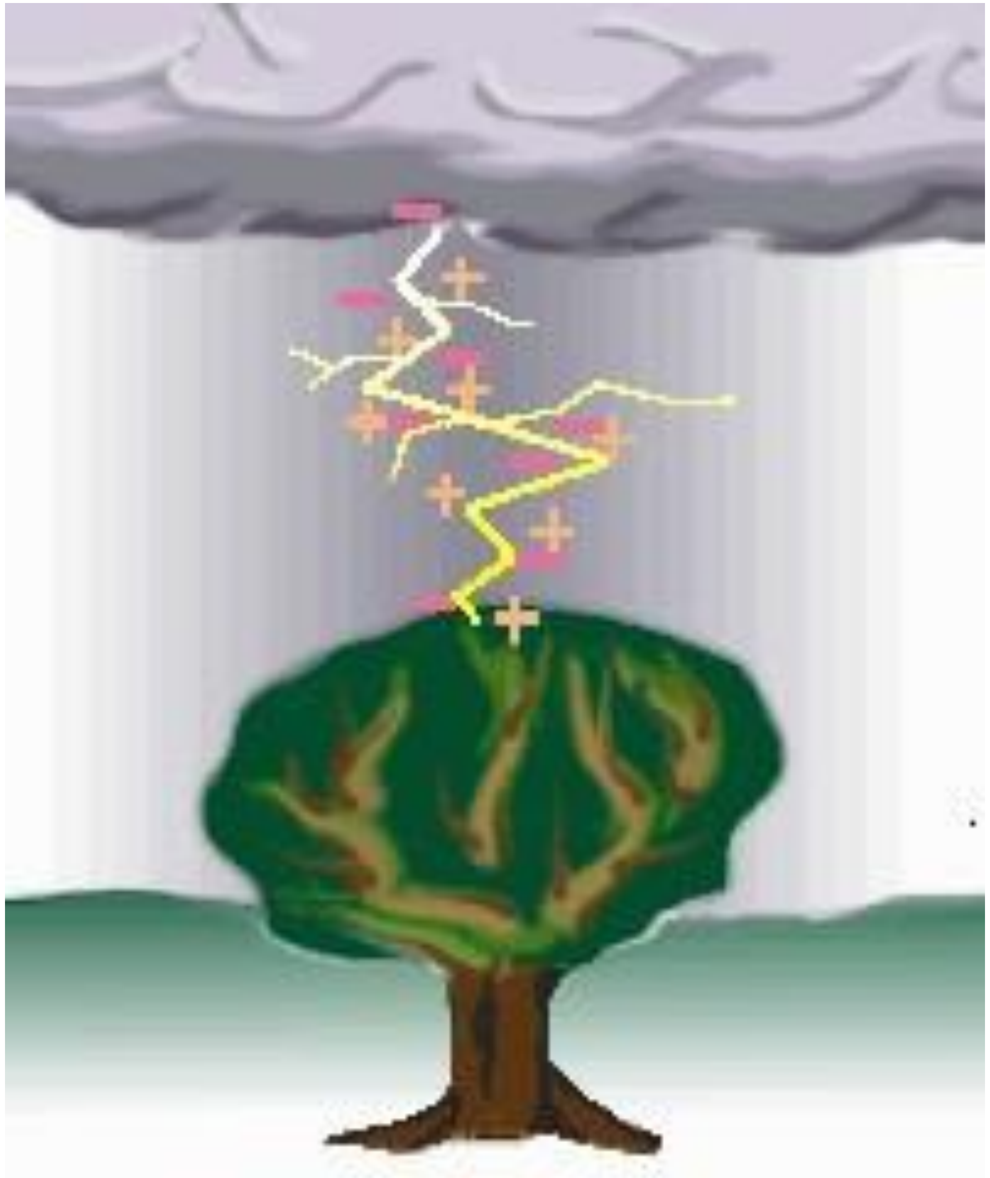
  ✓Cloud ↔ Cloud

  ✓Cloud ↔ Ground

  ✓Within a single cloud

How Lightning Occurs?


HOW LIGHTNING FORMS
WHEN THE CHANNELS CONNECT...



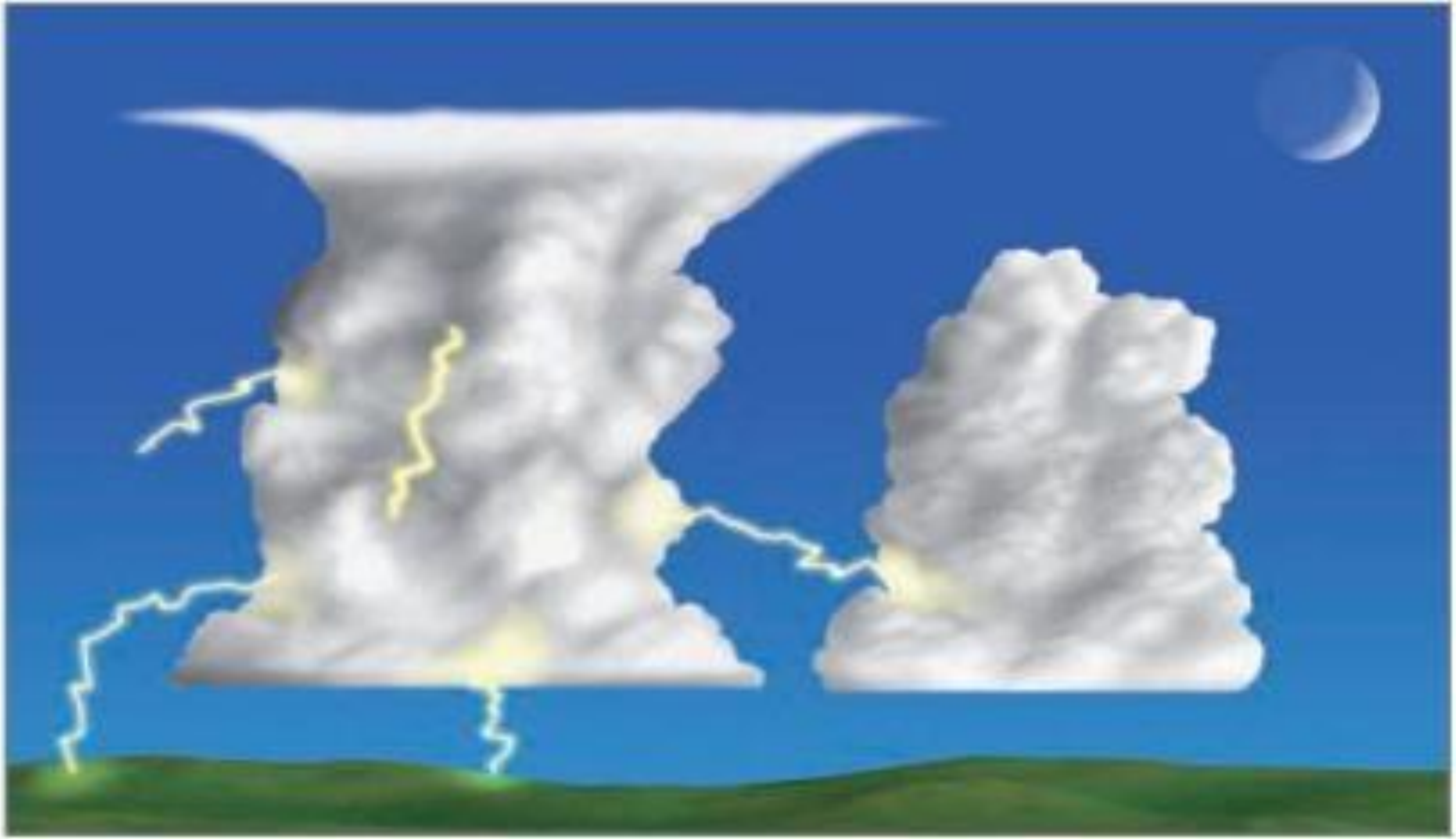
Conti……………

- Lightning is a discharge of electricity. A single stroke of lightning can heat the air around it to 30,000°C (54,000°F)!
- This extreme heating causes the air to expand explosively fast.
- The expansion creates a shock wave that turns into a booming sound wave, known as ***thunder.***

Conti……………

- Light travels so fast that we see light instantly after a lightning flash. But the sound of thunder, traveling at only about 330 m/sec (1100 ft/sec), takes much longer to reach the ear.
- If we start counting seconds from the moment we see the lightning until we hear the thunder, we can determine how far away the stroke is
- Because it takes sound about 3 seconds to travel 1 kilometer (5 seconds for each mile), if we see lightning and hear the thunder 15 seconds later, the lightning stroke occurred 5 km (3 mi) away.

Types of Lightning



Lightning Facts

- The energy from one lightning flash could light a 100-watt light bulb for more than 3 months.
- Most lightning fatalities and injuries occur when people are caught outdoors in the summer months during the afternoon and evening.
- Lightning can occur from cloud-to-cloud, within a cloud, cloud-to ground, or cloud-to-air.
- Many fires in the western United States and Alaska are started by lightning.

….END

Thanks !

Tropical Meteorology

Chapter 3: Tropical Weather Systems and Variability

Objective

3.2 Tropical Variability

- *To Define Madden Julian Oscillation*
- *To Know The Quise Biennial Oscillation*
- *To See the ElNińo-Southern Oscillation*

The causes of climate variability

- The climate variability of the tropical atmosphere is attributed to the variations in the *sea surface temperature* and it appears to be the primary contributor for the Northern Hemisphere winter climate.

- Variability generated by the interactions between the atmosphere, oceans and other earth systems is referred to as *coupled climate variability.*

- Climate variability can also be externally forced by *volcanic eruptions*, *variations in solar emission of radiation* and *anthropogenic causes* that change the atmospheric composition by green house gas emissions.

Conti……………

- Studies on global warming and climate change generally distinguish between *internally generated* and *externally-forced climate variability*.
- Internally generated variability is the result of processes *within the system*, while externally-forced variability is caused by some factors *outside the system.*
- A classic example of externally-forced climate variability is represented by the changes caused by variations in the amount and distribution of *solar energy* incident on the Earth because of the differences in the solar luminosity or in the Earth's orbital parameters.

Conti……………

- Internal climate variability not forced by external agents generally occurs on short time-scales of few weeks.
- Climate is capable of producing significant impact of internal variations of considerable magnitude without any external influences.
- These internally caused climate variability in the earth- atmosphere system are Madden Julien Oscillations (MJO), ElNino-Southern Oscillation (ENSO), QuasiBienniel Oscillation (QBO), Pacific Decadel oscillation (PDO) & North Atlantic Oscillation (NAO).

Madden Julian Oscillation (MJO)

Conti……………

- The MJO is the largest element of the intra-seasonal (30–60 days) variability in the tropical atmosphere.
- MJO is a major feature of the tropical circulation.
- This was discovered by Roland Madden and Paul Julian in 1971.
- It is a large-scale coupling between atmospheric circulation and tropical deep convection.

Conti……………

- It is a traveling pattern, propagating eastwards at approximately 4 to 8 m/s, through the atmosphere above the warm parts of the Indian and Pacific oceans.
- This overall circulation pattern manifests itself in various ways, most clearly as anomalous rainfall.

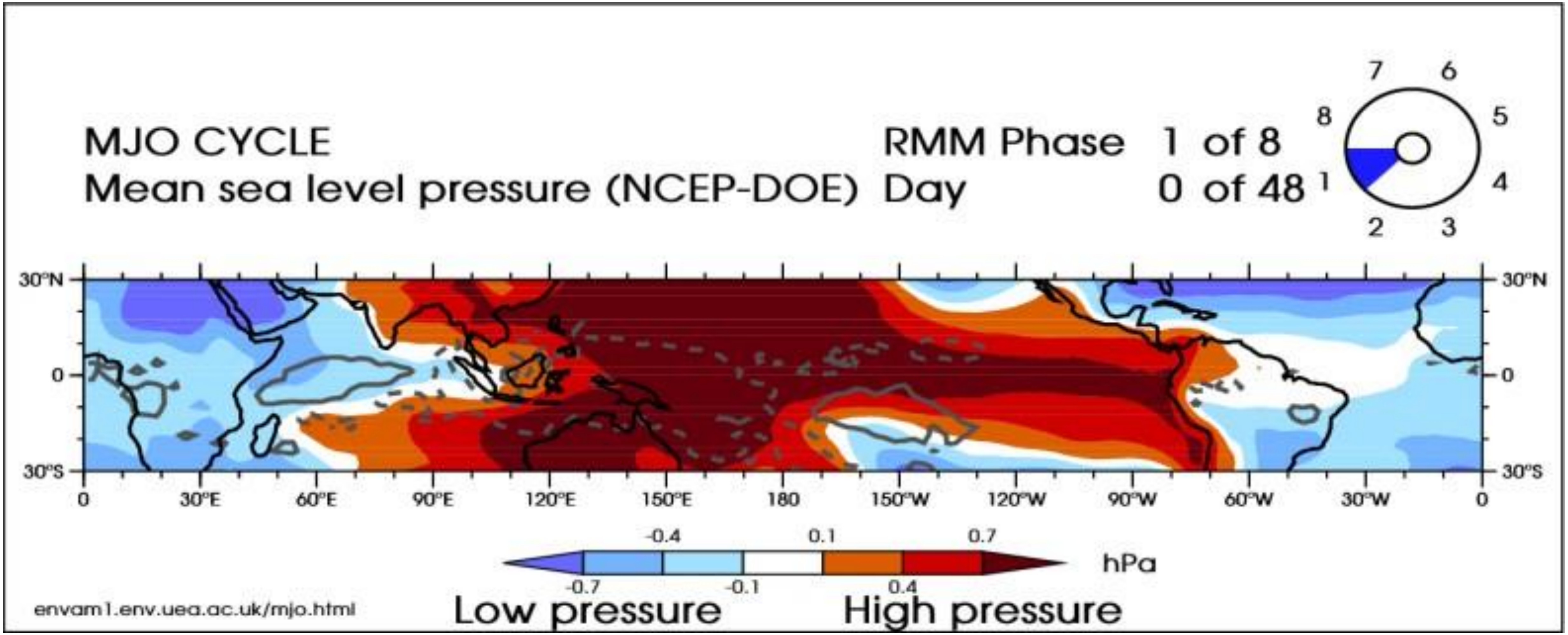
Conti……………

- The MJO is characterised by an *eastward* propagation of rainfall over the 'warm pool' region from the western tropical Indian Ocean to the Central Pacific.
- The anomalous rainfall is usually first evident over the western Indian Ocean, and remains evident as it propagates over the warm ocean waters of the western and central tropical Pacific.
- This pattern of tropical rainfall then generally vanishes as it moves over the cooler ocean waters of the eastern Pacific.

Conti……………

- The MJO cycle is split up into 8 phases, for convenience.
- Each phase corresponds to 1/8 of the full cycle. An individual MJO event can last anywhere between 30 and 60 days (the MJO is a 'broadband' phenomenon).

Conti……………

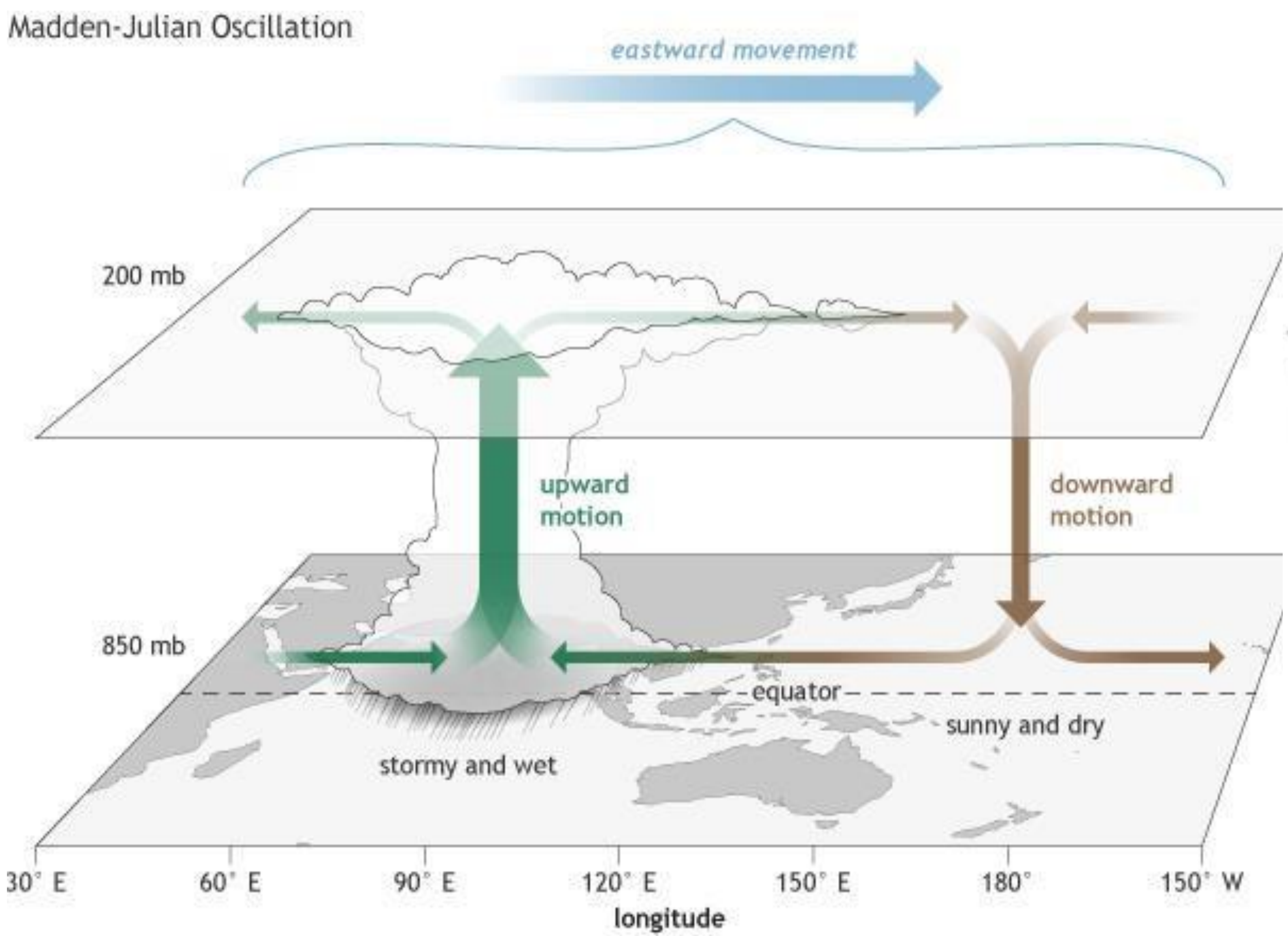


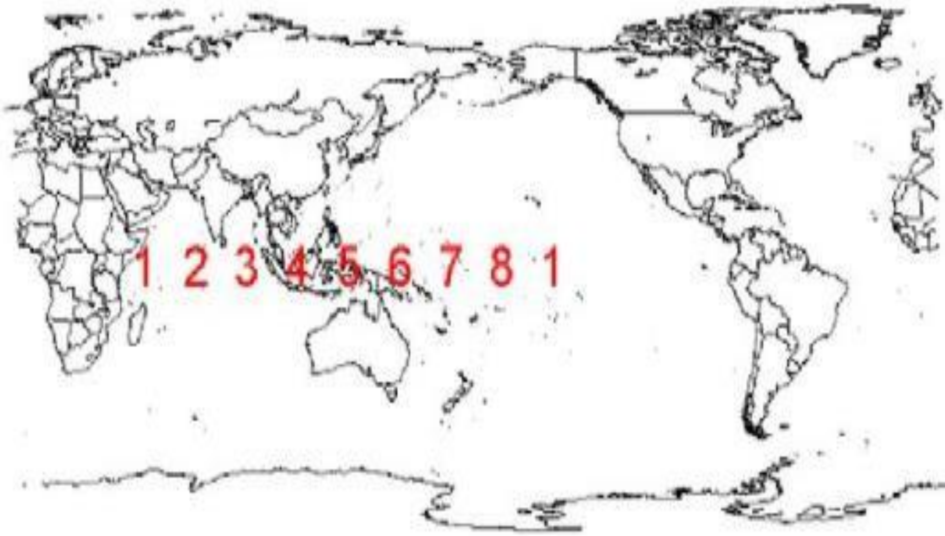
MJO CYCLE
Mean sea level pressure (NCEP-DOE)
RMM Phase 1 of 8
Day 0 of 48
7 6 8 5 1 4 2 3
30°N
0
30°S
0 30°E 60°E 90°E 120°E 150°E 180 150°W 120°W 90°W 60°W 30°W 0
-0.4 0.1 0.7
-0.7 -0.1 0.4
hPa
envam1.env.uea.ac.uk/mjo.html
Low pressure
High pressure

Conti…………….

- The MJO consists of two parts, or **phases**: one is the enhanced rainfall (or **convective**) phase and the other is the suppressed rainfall phase.
- Strong MJO activity often dissects the planet into halves: one half within the enhanced convective phase and the other half in the suppressed convective phase.
- These two phases produce opposite changes in clouds and rainfall and this entire **dipole** (i.e., having two main opposing centers of action) propagates eastward. The location of the convective phases are often grouped into geographically based stages that climate scientists number 1-8 as shown in Figure 1.

Conti……………


Madden-Julian Oscillation
eastward movement
200 mb
850 mb
upward motion
downward motion
equator
sunny and dry
stormy and wet
30° E
60° E
90° E
120° E
150° E
180°
150° W
longitude


1 2 3 4 5 6 7 8 1

Effects of MJO

Conti……………

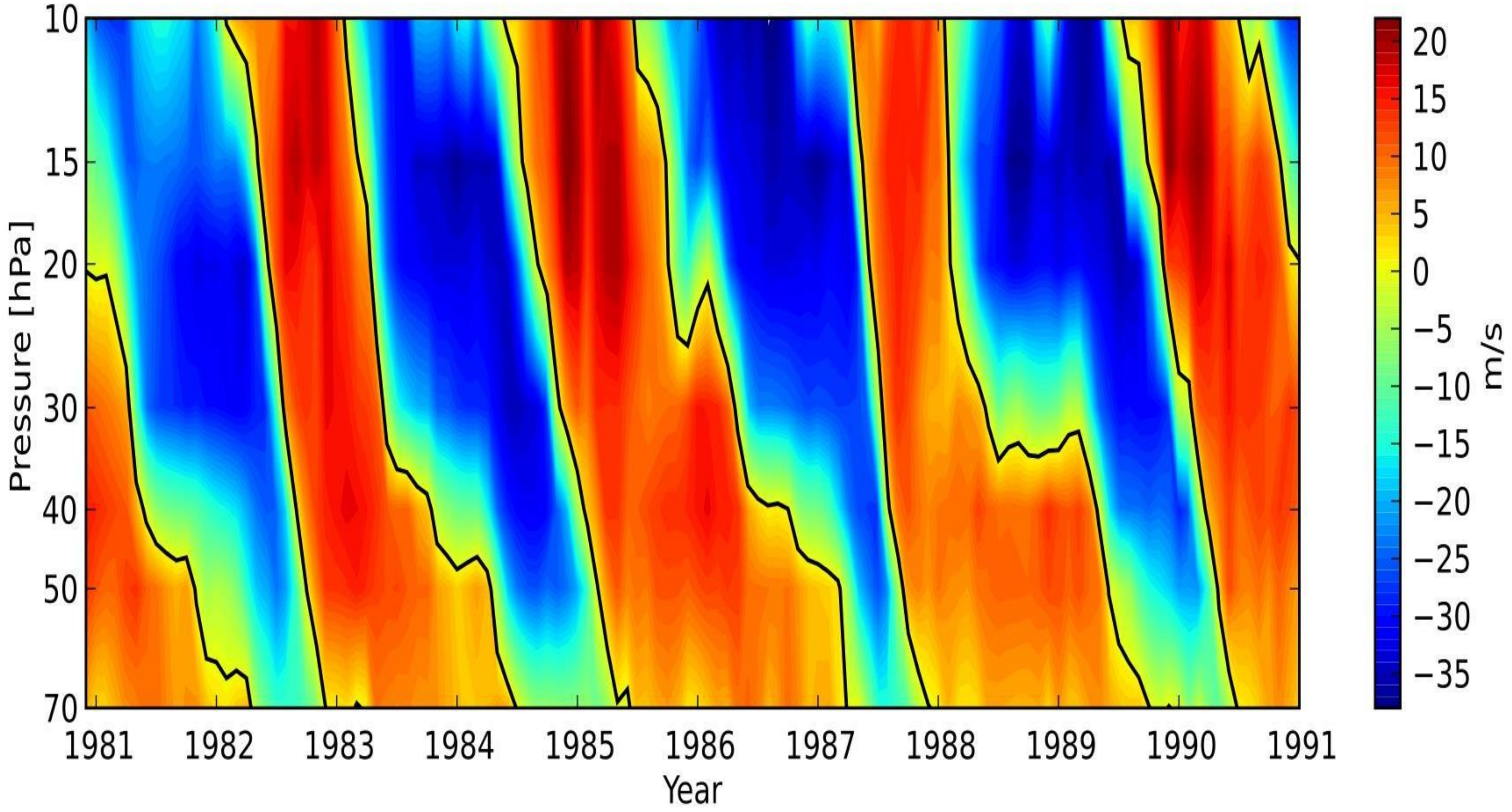
- The MJO also affects other meteorological systems in the tropics like monsoons, tropical cyclones or hurricanes and ENSO.
- *Monsoons:* The MJO modulates the active/break cycles that occurs within the Asian and West African monsoons.
- *Tropical cyclones/hurricanes:* The MJO reduces tropical cyclone numbers.
- *El Nino - Southern Oscillation (ENSO):* The MJO has been influential in causing recent El Nino episodes.

Quise Biennial Oscillation (QBO)

Conti................

GENERAL CHARACTERSTICS OF QBO

- The periodicity of the oscillations is 28 months (varies between 20 & 32 months).
- Mean zonal winds in the stratosphere near 30 hpa level between 30°n and 30°s show alternating easterly and westerly regimes.
- Easterly phase is prevalent during odd years and westerly during even years.
- Its max amplitude is at 30 hpa and over the equator.
- It decreases downward and poleward.
- It is hardly noticeable below 100 hpa.


10
15
20
30
40
50
70
Pressure [hPa]
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
Year
20
15
10
5
0
−5
−10
−15
−20
−25
−30
−35
m/s

The QBO has the following observed features:

- Zonally symmetric easterly and westerly wind regimes alternate regularly with periods varying from about 24 to 32 months with an average *of 28 months*.
- Successive regimes first appear above 30 km, but propagate downward at a rate of 1 km per month.
- The downward propagation occurs without loss of amplitude between 30 and 23 km, but there is rapid attenuation below 23 km.
- The oscillation is symmetric about the equator with maximum amplitude of about 20 m s-1, and an approximate Gaussian distribution in latitude with a half-width of about 120.

Conti……………..

- A remarkable feature of winds near the equator at about 25 km elevation (where pressures are about 25 hPa) is their reversal between comparatively fast easterlies and slow westerlies and then back again, taking about two years. So it is called the *Quasi-Biennial Oscillation* (QBO).
- For instance, the easterly winds over Singapore reached a maximum of around 15 m/ s in 1982, 1985, 1987 and 1990, i.e. there is a cycle of variation of about 26 months.

Conti…………….

- The QBO is excited by vertically propagating long period waves like Rossby and Kelvin waves which causes westerly accelerations in westerly shear zones and easterly accelerations in easterly shear zones.
- The Quasi-biennial Oscillation (QBO) is primarily associated with fluctuations in atmospheric layer of stratosphere.

Effects of QBO

Conti……………..

- Effects of the QBO include mixing of stratospheric ozone by the secondary circulation caused by the QBO, *modification of monsoon precipitation*, and *an influence on stratospheric circulation in northern hemisphere winter* (mediated partly by a change in the frequency of sudden stratospheric warmings).

- Easterly phases of the QBO often coincide with more sudden stratospheric warmings, *a weaker Atlantic jet stream* and cold winters in Northern Europe and eastern USA whereas westerly phases of the QBO often coincide with mild winters in eastern USA and a strong Atlantic jet stream with mild, wet stormy winters in northern Europe.

Conti……………

- In addition, the QBO has been shown to affect hurricane frequency during hurricane seasons in the Atlantic.

ElNino-Southern Oscillation (ENSO)

Conti……………..

- *El Nino* is the name given to the appearance of anomalously warm surface water off the South American coast, a condition which leads periodically to catastrophic downturns in the Peruvian fishing industry by severely reducing the fish catch that severely affects the economy of the country.
- The colder water that normally upwells along the Peruvian coast is rich in nutrients, in contrast to the warmer surface waters during ElNino.

Conti……………

- The phenomenon has been linked by Bjerknes (1969) with the Southern Oscillation as some kind of air-sea interaction effect called ENSO.
- In fact, El Niño and the Southern Oscillation are simply aspects of the same global *episode*, where this title ENSO is compounded from ElNiño+Southern Oscillation.

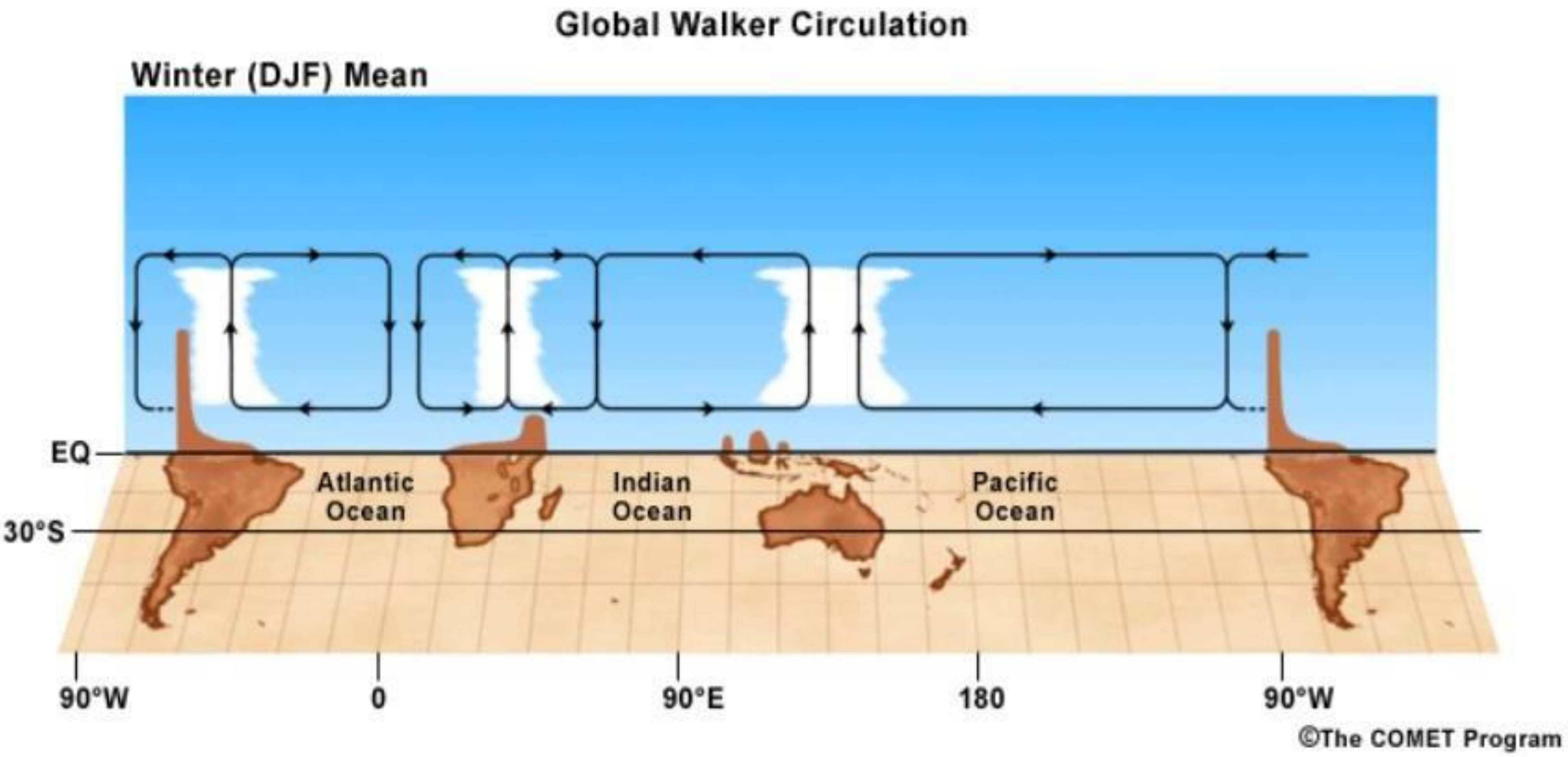
Conti……………

- ENSO is a coupled atmospheric-oceanic process caused by the oscillatory redistribution of heat and atmospheric momentum in the equatorial Pacific Ocean.
- The term "coupled" means that the atmosphere and ocean work in harmony with one another; forcing in one modulates the other and vice versa.
- ENSO has an irregular, two to seven-year, cycle and large areal extent (the entire tropical Pacific), and its impacts are global.

Walker Circulation

- There is a pattern of zonal flows over the equator across the Pacific Ocean named the *Walker circulation* after Gilbert Walker, who discovered it in the 1920s.
- The strength and direction of the circulation are measured by the difference between sea-level pressures (in hPa) at Papeete (17°S in Tahiti in the central Pacific) and Darwin (12°S in northern Australia), 8,500 km away.
- Tahiti is under the influence of the South Pacific high and experiences easterly Trades most of the year, while Darwin is near the west equatorial Pacific warm pool and has a distinctly monsoonal climate.

Conti……………



Global Walker Circulation
Winter (DJF) Mean
EQ
30°S
Atlantic Ocean
Indian Ocean
Pacific Ocean
90°W
0
90°E
180
90°W
©The COMET Program

ElNino

- Under normal conditions the cold Humboldt (or Peru) current flowing along the coast of Peru & Ecuador (Latin America) gives strong upwelling and so fish production is very high along this coast. This is called “La Nina” meaning ‘a girl child’.
- But under abnormal conditions, this narrow and cold Humboldt current along west coast of South America is replaced by warm water advection off the coast of Peru & Ecuador. This usually happens during Christmas time of December. So it was named ‘ElNino’ which means “the child Christ” in Spanish.
- El Niño and La Niña refer the oceanic component of ENSO and reflect modulations in the strength of the easterly trades and pattern of sea surface temperatures across the equatorial Pacific Ocean.

Conti……………..

- The Southern Oscillation refers to the atmospheric component of ENSO and reflects shifts in patterns of sea level pressure across the tropical Pacific that occur in conjunction with El Niño (warm) and La Niña (cold) ENSO phases.
- *El Niño* is characterized by weakened or reversed tropical easterly trades and the concurrent warming of equatorial sea surface and below-sea surface temperatures across the eastern and central tropical Pacific.
- Conversely, La Niña is characterized by strengthened easterly trade winds and the concurrent cooling of equatorial sea surface and belowsea surface temperatures across the eastern and central tropical Pacific.

Conti……………

- El Niño, is characterized by weakening of the trade winds and warming of SSTs in the Equatorial Pacific.
- El Niño or "Christ Child" originally referred to the appearance of warm water off the coast of Peru and Ecuador near Christmas time.

Conti……………..

- It was noticed that in some years, the warming was stronger than others and disrupted local fishing.
- La Niña is associated with stronger than normal trade winds and the anomalously cool SST.
- On average, La Niña is a less extreme anomaly than El Niño but tend to last longer, 1-3 years.

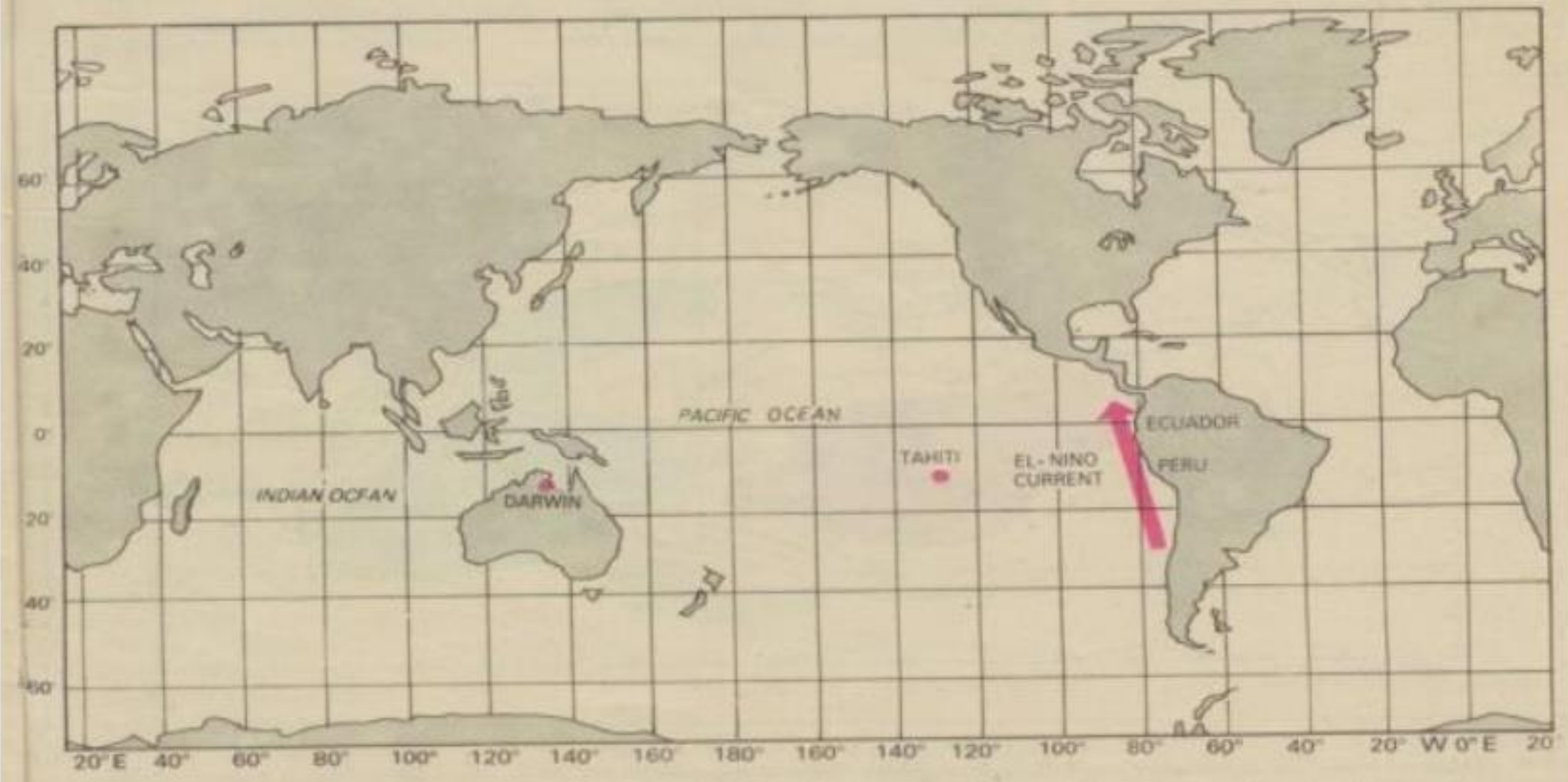


Fig. 2. The El Nino and Southern Oscillation Index (SOI). SOI=Tahiti minus Port Darwin pressure

Southern Oscillation

- The Southern Oscillation is a phenomenon manifesting as a seesaw pattern in the sea level pressure between the Indian Ocean (Darwin, in northern Australia) and the Eastern Pacific Ocean (Tahiti, in French Polynesia).
- This oscillation has a time scale of roughly 4–6 years and amplitude on the order of a few millibars.
- It bears the name of the Southern Oscillation due to the fact that its largest amplitudes are found in the Southern Hemisphere, between the Equator and 50 degree S.

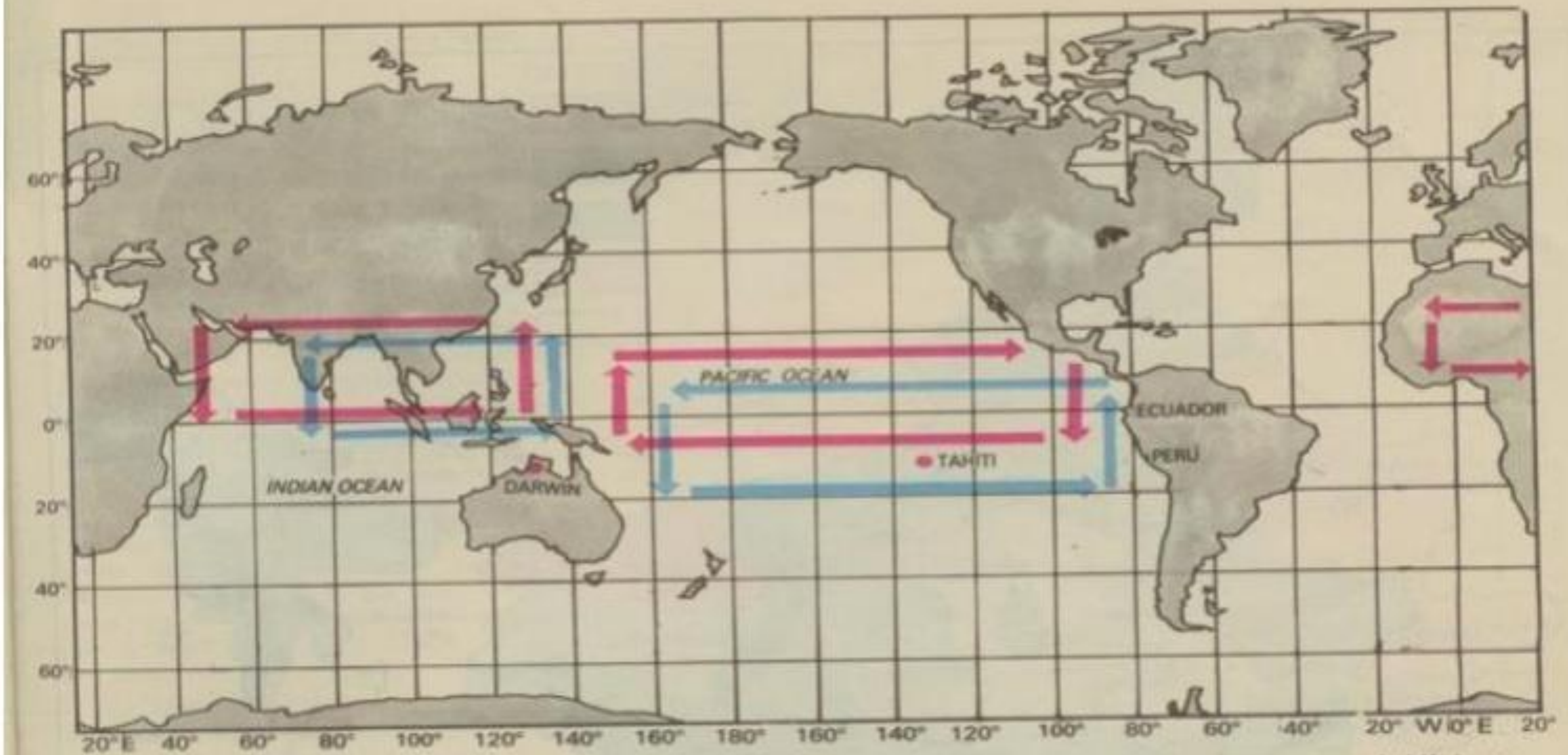
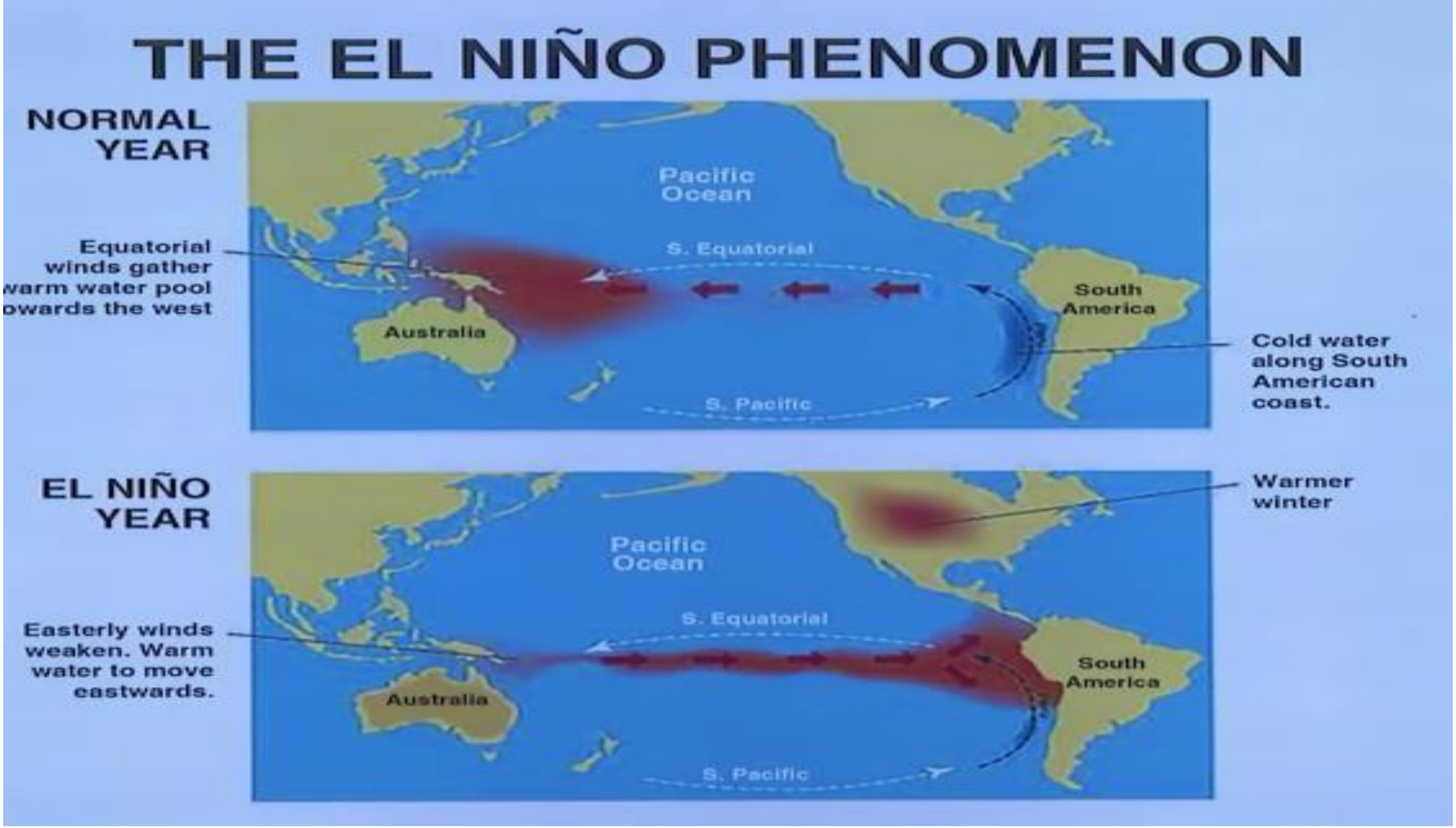
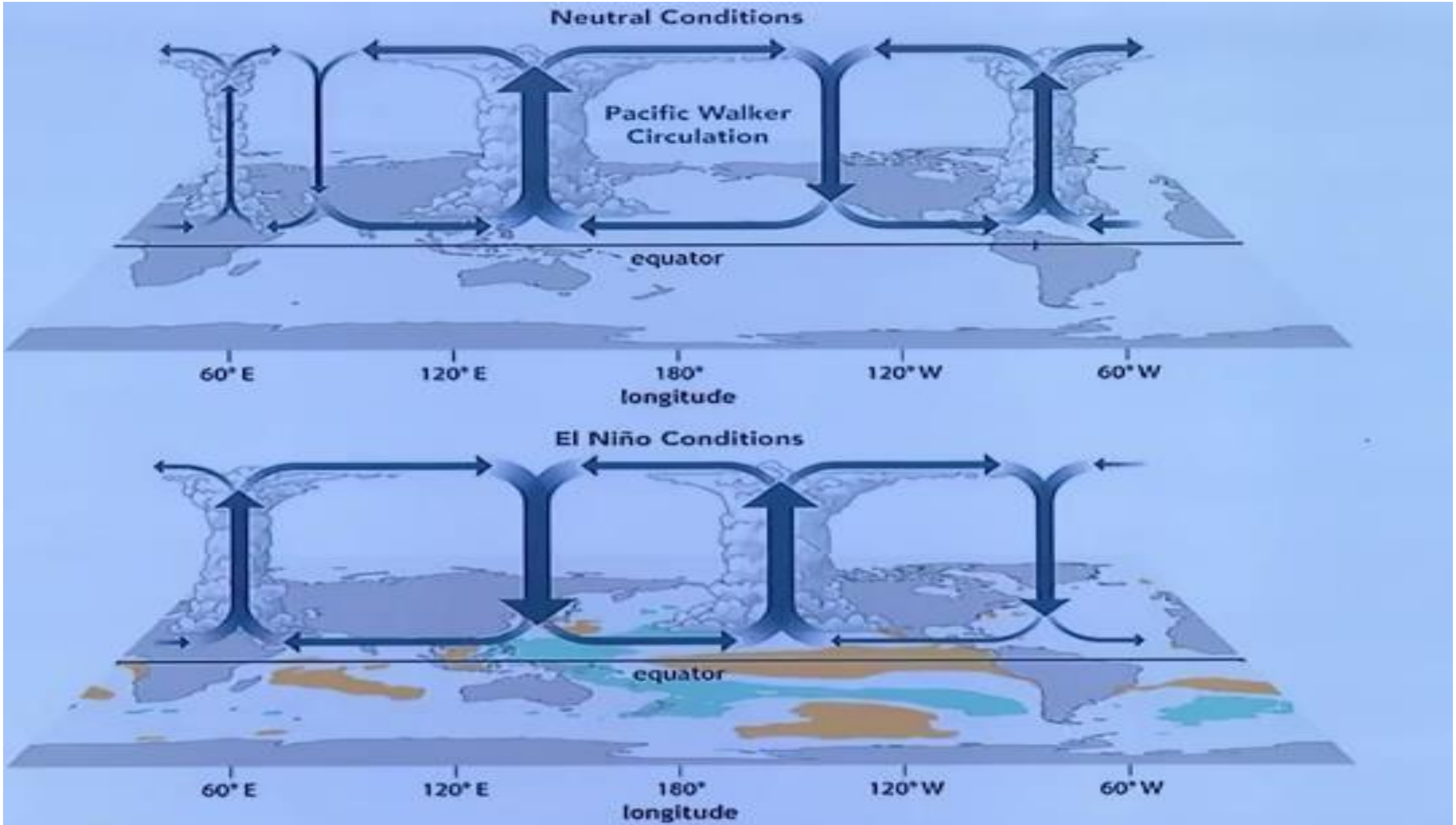
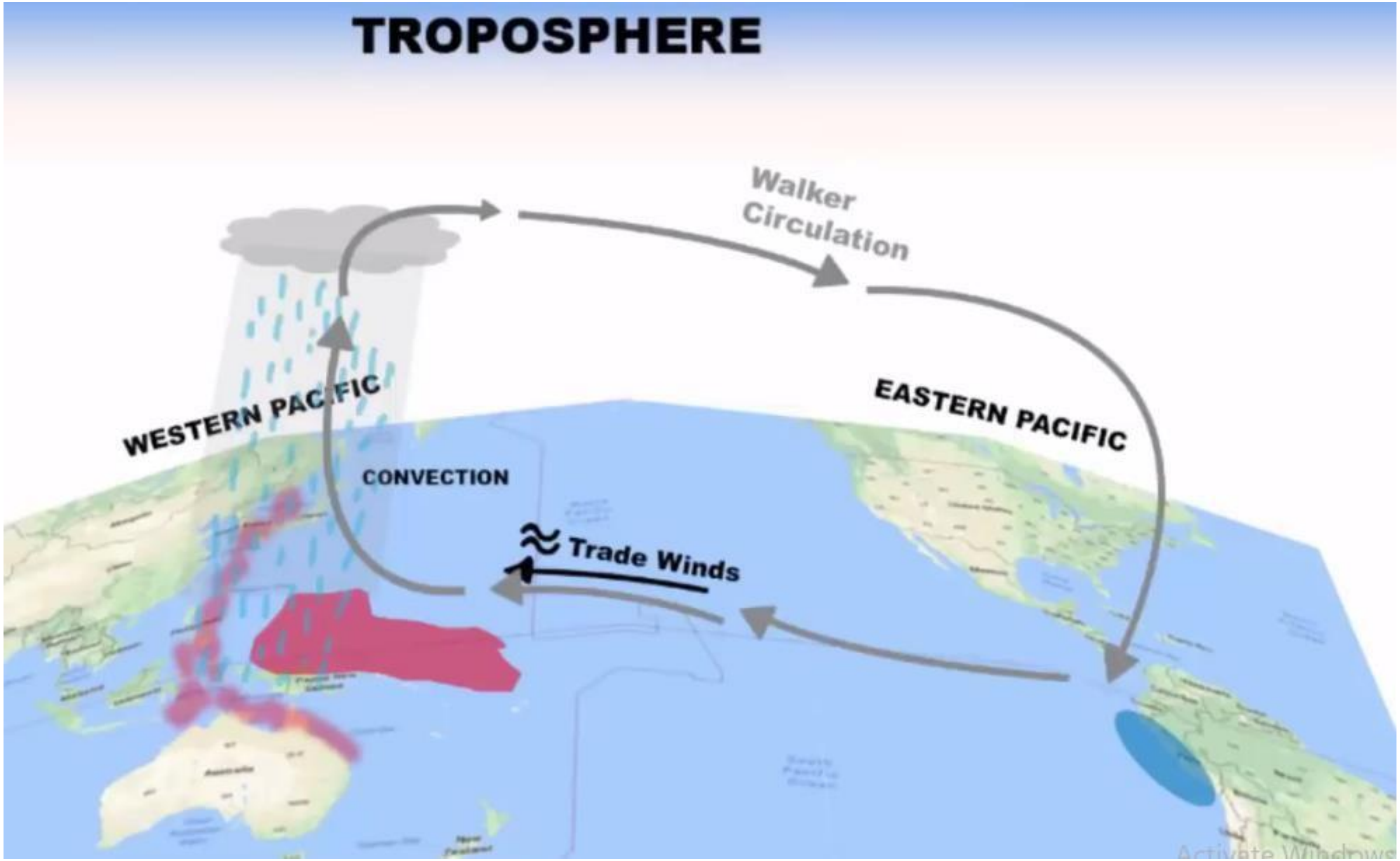
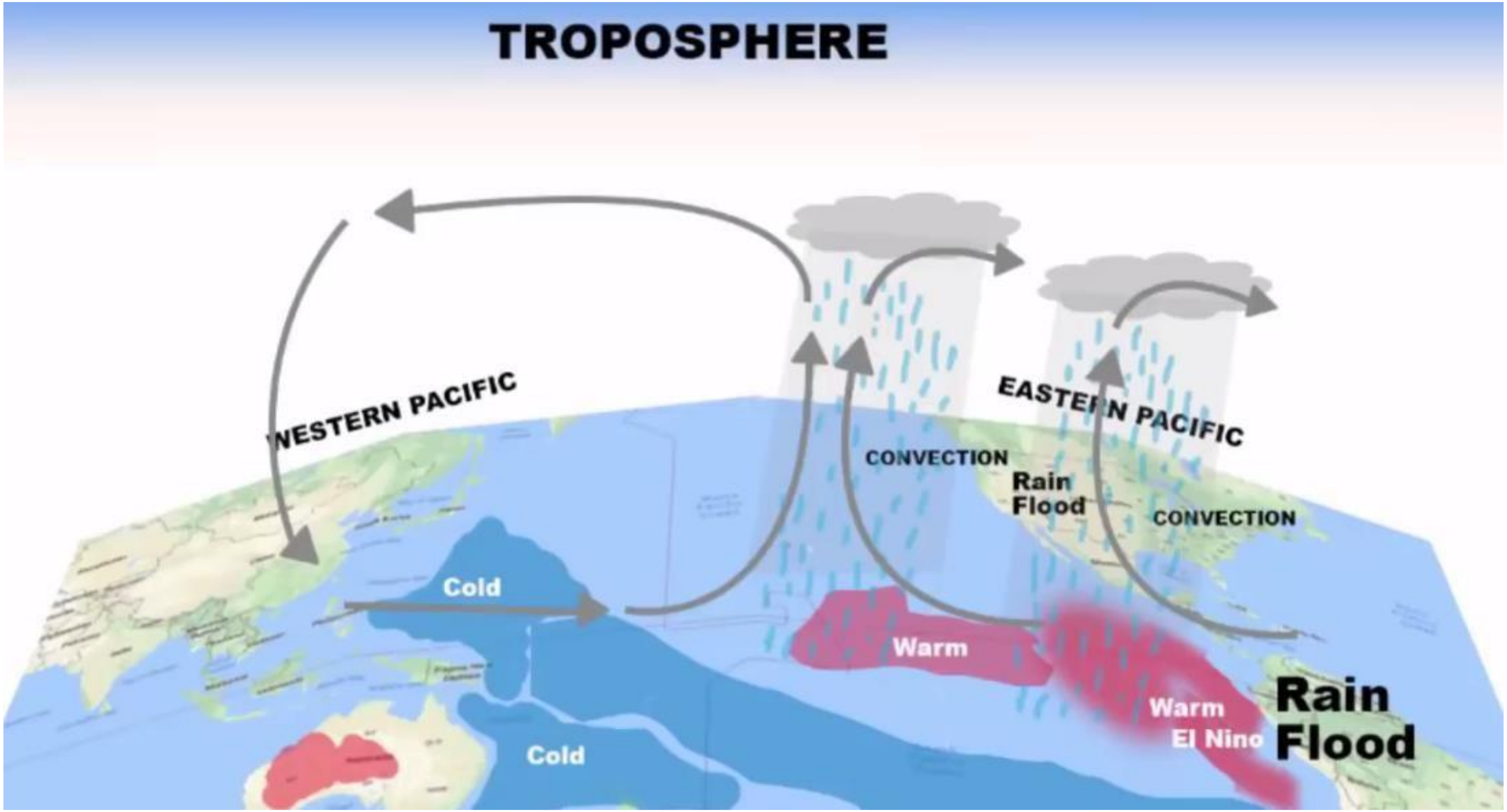


Fig. 4. Walker cells. Red arrows indicate normal year. Blue lines indicate abnormal years. Note the eastward shift in an


THE EL NIÑO PHENOMENON
NORMAL YEAR
Equatorial winds gather warm water pool owards the west
Pacific Ocean
S. Equatorial
South America
Australia
Cold water along South American coast.
S. Pacific
EL NIÑO YEAR
Warmer winter
Pacific Ocean
S. Equatorial
Easterly winds weaken. Warm water to move eastwards.
South America
Australia
S. Pacific


Neutral Conditions
Pacific Walker Circulation
equator
60° E
120° E
180°
120° W
60° W
longitude
El Niño Conditions
equator
60° E
120° E
180°
120° W
60° W
longitude


TROPOSPHERE
Walker Circulation
WESTERN PACIFIC
EASTERN PACIFIC
CONVECTION
Trade Winds


TROPOSPHERE
WESTERN PACIFIC
EASTERN PACIFIC
CONVECTION
Rain
Flood
CONVECTION
Cold
Cold
Warm
Warm
El Nino
Rain
Flood

Conti……………..

- The Southern Oscillation was first substantiated by Sir Gilbert Walker, who noticed an oscillatory pattern within the sea level pressure differences between Darwin, Australia and Tahiti while attempting to forecast the Asian and Australian monsoons.
- Southern Oscillation Index (SOI) is the normalized measure of this difference in sea level pressure which is one of the primary means of monitoring the state and evolution of ENSO.

Conti……………

- SOI is expressed by $$SOI = \frac{10(p_{diff} - \overline{p_{diff}})}{\sigma(\overline{p_{diff}})}$$

Where $p_{diff} = p(\text{Tahiti}) - p(\text{Darwin})$

p_{diff} is the difference in monthly mean sea level pressure between Tahiti and Darwin.

$\bar{p}_{diff}$ is the difference in monthly mean sea level pressure between Tahiti and Darwin.

- The SOI is a proxy for the strength of trade winds as pressure differences wind speed.
- Negative or low SOI represents EL Nino conditions while positive or high reflect La Nina.

Table 1

(a) Positive SOI:	(i)	Tahiti pressure greater than Port Darwin.
	(ii)	Pressures high over east Pacific and low over Indian Ocean.
	(iii)	Low rainfall over eastern Pacific and prospects of good monsoon rain over India and Indian Ocean.
(b) Negative SOI:	(i)	Port Darwin pressure exceeds Tahiti.
	(ii)	Pressures high over Indian Ocean and low over eastern Pacific.
	(iii)	Low rainfall or poor monsoon over Indian Ocean and higher than usual rain over east Pacific.

….END

Thanks !